

# Building Assessment Report

PPS One North Commercial Building Inspection
2025-1106

**Prepared for:**

Portland Public Schools

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# Architectural

## Executive Summary

### Why One North?

One North Center, an iconic building development in the historic Albina neighborhood of northeast Portland, provides prime inspirational and creative space for Portland Public School's Center for Black Student Excellence (CBSE) program. The existing buildings and site exemplify forward and future thinking in collaboration and community since its construction in 2015.

### Location

**One North offers a uniquely strategic and inspiring setting for the PPS CBSE program.** Located on North Fremont Street between North Williams and North Vancouver Avenues, the site serves as an established central and accessible hub for neighborhood schools, commercial businesses, and supporting programs. The urban city development consists of two commercial office buildings with street-level courtyards.

| | Street Address | Stories | Lot size (Parcels) |
|---|---|---|---|
| Karuna II East w/East Courtyard | 25 N Fremont Street | 4 story | ~153 ft x 118 ft (3) |
| Karuna II West w/West Courtyard | 3506 N Vancouver Avenue | 5 story | ~103.2 ft x 120 ft (2) |
| Public Alley | Fire lane | | 300 ft x 20 ft (1) |




*Figure A-1 Aerial Site view*

# Architectural Summary

This section summarizes the architectural assessment of the Karuna II West and East buildings at One North, conducted for Portland Public Schools (PPS) to support the Center for Black Student Excellence (CBSE). The evaluation covers accessibility, code compliance, safety, and maintenance concerns and recommendations.

The Karuna II West and East buildings were designed and constructed at the same time with similar materials and finishes. Because of these similarities, the report addresses the site, building exteriors and interiors together and differences noted. Observations and recommendations within this report were assembled from information gathered during two site visits and review of information provided from the building Owner's representative. A general summary of recommendations is included here with additional conditions and recommendations provided in the following sections.

## Exterior and Site Conditions

- Maintain wood veneer, sealants, and roof flashings proactively.
- Repair uneven courtyard paving system, particularly at paths along building entries/ exits.
- Evaluate bicycle parking for zoning compliance.

## Accessibility and Code Compliance

- Measure exit path illumination levels to ensure code compliance.
- Upgrade signage for accessible exits and elevator landings.
- Install van-accessible parking signage and pavement markings per Oregon code.
- Relocate toilet accessories that interfere with ADA grab bar clearances.
- Replace protruding dispensers exceeding 4 inches along accessible routes.

## Safety and Egress

- Improve courtyard lighting levels for visibility during dusk to dawn hours.
- Review building code appeals for stair rail termination issues.
- Repair loose stair base trim and evaluate tight stair landing conditions.

## Roof and Terrace Maintenance

- Install and secure roof drain strainers where missing.
- Improve roof walkway access around solar panels for code compliance.
- Repair sagging mechanical screen enclosure gates.
- Verify terrace door opening forces meet accessibility standards.
- Consider additional fall protection measures with PPS standards.

# Architectural Assessment

The Karuna II East and West buildings are located within a Commercial Mixed Use 3 (CM3) zone, surrounded on three sides by public transit and bikeway streets—unlike traditional District campuses that typically feature a single building enclosed by fencing. This urban setting supports walkability, multimodal access, and community integration.

The East and West buildings are on separate properties with an alley between. The neighboring properties to the north share use of the alley. This alley is dedicated fire access for both East and West buildings and their northern neighbors. The open space between buildings, including the fire lane, comprises the open shared courtyard amenity space in common to the 4 properties.

City zoning regulations do not require on-site parking. Service loading on adjacent streets requires advance approval from the city. There is limited on-site parking on the Karuna II East property with eight covered parking spaces accessed from North Williams Avenue. These spaces are reserved and monitored with towing risk. During site visits, café patrons and delivery vehicles actively used the central open space.

The open courtyard design—a key feature of the city approved *"woonerf"* (pedestrian street) concept—appears to embody the CBSE strategic goals of community engagement, collaboration, and innovation. While this open access supports shared use and flexibility, it contrasts with PPS's typical safety standards, which rely on gates, bollards, and fencing to control vehicular access. The District is encouraged to explore context-sensitive safety and security solutions that respect the urban character of the site while ensuring student and staff safety.




*Figure A-2 Site diagram – West and East buildings, Fire Lane, and Courtyard*




*Figure A-3 One North - Directional sign for vehicular parking*

## Site and Courtyard Conditions

**Street frontages on the east, south and west with public sidewalks provide the boundaries for the development.** The public sidewalks were clean and evenly surfaced in good condition. Radiused corners at intersections featured low slope curb ramps with yellow truncated dome detectable warning pads free from uneven joints and appear in good condition.

**Sidewalk planters located along the street curb are well spaced and planted with mature trees, grasses, and mulch.** Additional raised planters are part of a painted steel shade structure on busy N Fremont Street. The structure is 3-stories high and approximately 24 inches deep buffering pedestrians from the building interior along the storefront glazed building street walls.

At the property line, paving materials change from the light-colored concrete to dark stone pavers continuous to the buildings entries and through the courtyard. In the same character as the street planters, the courtyard features larger planters with mixture of plants, tall mature trees, grasses, mulch, and irrigation.

**Plant and paving materials are consistent across the site.** They appear well established and well maintained with areas in need of plant renewal. Low steel rails stamped with artful comments line planter edges at several locations. The color contrast and height are low, presenting tripping hazards where paths intersect.

**The site slopes down from northeast to southwest.** Connective walking paths are low slope and appear to be accessible routes. Planters and low concrete faced retaining walls provide structured height transitions – notably at the Karuna II East building courtyard. Steel punched plates at planters' edge and large organic shaped benches mark the boundaries for pedestrians and vehicles to navigate within the courtyard. Removable bollards at the north end of fire lane limit traffic from driving through. Other fixtures on site include bicycle racks near building entry doors, benches, and bollard lighting. Irrigation heads were visible within courtyard planting beds.

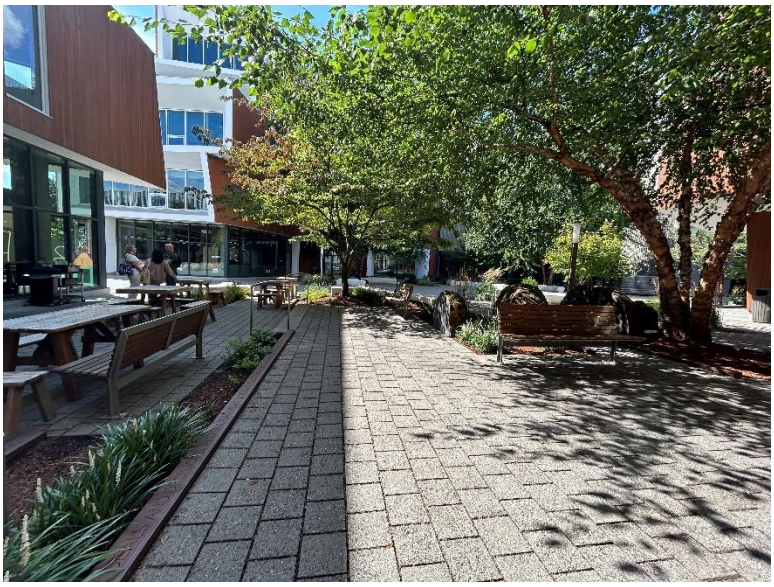


*Figure A-4 Karuna II East Courtyard view to west*



*Figure A-5 Karuna II East Courtyard view to north*

**Short term bicycling parking is provided near building entry doors with sidewalk and courtyard racks.** Long term bicycle parking in both the East and West buildings, when constructed, featured wall mounted bicycle racks lining the lobbies as an innovative approach to visibility, access, security, and sustainability. This conformed with City zoning long term bicycle parking standards at the time of construction. The East and West lobby fixtures vary from the construction documents. No bicycle racks were currently in place in the West building lobby. Wall mounted racks at angled walls in the East building lobby appeared as protruding objects when entering from the courtyard lobby door.

**Area for loading services and car parking is limited on the site.** The northeastern neighboring property utilizes the central courtyard's dedicated fire lane for access to its car parking area. The Owner's Representative noted that requests to secure the perimeter entry/exit points and parking areas of the site with removeable bollards, fencing, and gates to limit vehicular traffic into the courtyard have been unsuccessful with the city. Our assumption is that this is due to the fire lane access requirements and adjacent property owner access agreements and defies the spirit of the initial design as a 'woonerf' aka pedestrian street. Signage is the only available designator of ownership at present. During our visit, vehicles were observed utilizing the fire lane access and parking in areas clearly marked "no parking."

**Each building has a waste collection room contained within the building.** The locked rooms are shared among tenant spaces and fitted with a pair of swing doors for staff to maneuver waste collection, recycling, and the compost carts and bins out on collection days. This varies from typical District campus exterior collection areas accessed from a service drive. Karuna II East waste collection is accessed from the covered parking area through a pair of swing doors. Karuna II West collection has access from the main lobby interior with exterior doors that open south onto the N Vancouver sidewalk.



*Figure A-6 Karuna II East Van Accessible parking signage non-conforming*

### Recommendations

- Observed uneven surfaces and varying joint widths near the West building exterior exit passage and the north entry lobby call box. Other areas may exist, and further review is recommended to repair paver and material transitions resulting in smooth and level surface with well-anchored materials.
- Evaluate exterior lighting levels for conformance with exit path illumination code. include building courtyard exits and connecting paths within the courtyard to the fire lane.
- Coordinate waste collection pick up location and days with tenants and adjacent building owners. Plan for Facilities operational staff to maneuver carts to and from locations on service days.
- An evaluation of bicycle parking provided versus required for the building site is recommended to maintain conformance with city zoning codes as building improvements occur in time.

- Recommend including irrigation and planting, particularly the grass and tall trees, as part of a regular maintenance cycle. The grass and mulch replacement cycle will be more frequent than the irrigation and paving system.

## Site Accessibility

Visual survey during site visits found areas of improvement to bring accessibility and wayfinding signage into conformance with standards.

### Recommendations

- Recommend a detailed evaluation and develop conformance schedule for the buildings and site accessible signage.
- Karuna II East building accessible parking: Van Accessible sign (R7-8P) and Access Aisle pavement markings required per Oregon Revise Statue (ORS) 447.233 and Oregon Structural Specialty Code Chapter 11 were missing.
- Karuna II East and West buildings: Elevator landing and interior tenant signage indicating accessible exits.

## Site Safety and Egress

The south facing open-air courtyard welcomes pedestrians to enjoy its seating and café areas. It also invites service vehicles without limitation in contrast to the typical school sites where vehicular traffic is separated from student areas and controlled with bollards, gates, or fencing.

Site courtyard is direct access from N Fremont sidewalk and is not fenced off with a secure site perimeter. It was noted that the City of Portland will not allow bollards at driveway entrance from street, it must remain open access.




*Figure A-7 Karuna II East Parking and Main Accessible Lobby Entry*



*Figure A-8 Karuna II East Main Lobby and parking entry from Williams Street*

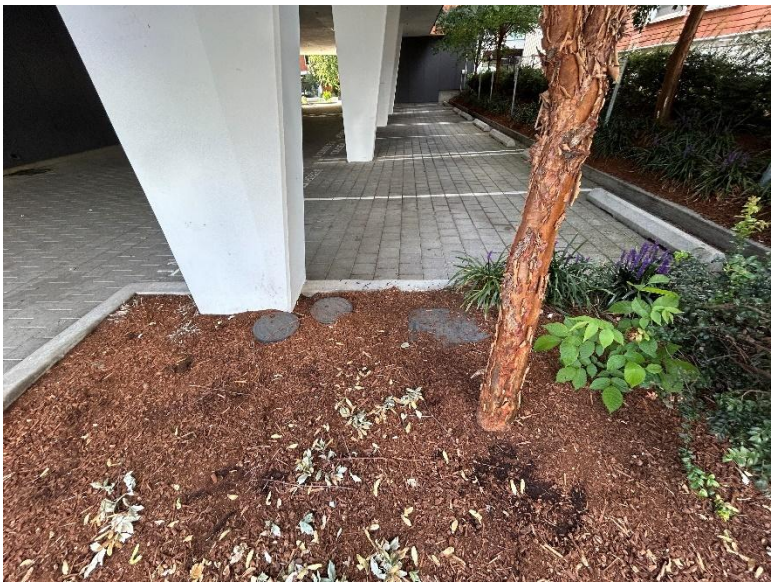


*Figure A-9 Karuna II East parking landscape buffer*

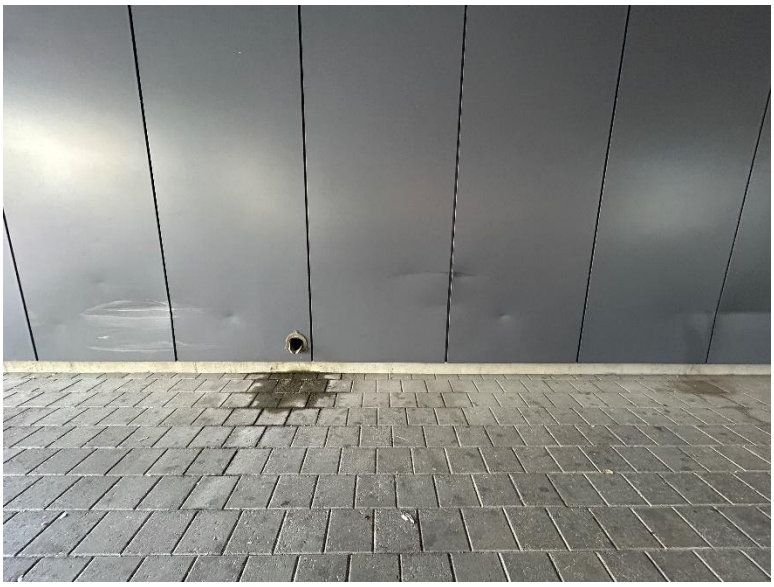


*Figure A-10 Karuna II East parking wall damage from cars*




*Figure A-11 Karuna II West Main lobby street entry.*



*Figure A-12 Karuna II West courtyard main lobby entry door.*
*Prefinished storefront window and door system.*
*Door bottom rail height is less than 10 inches.*

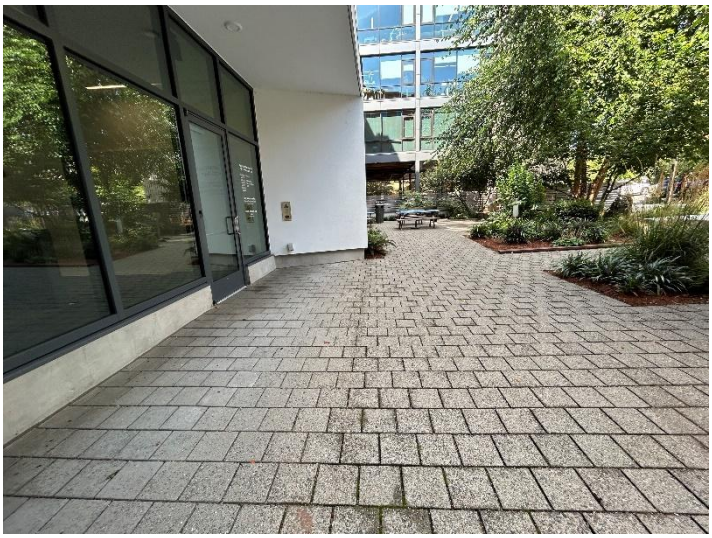


*Figure A-13 Karuna II West courtyard entry door approach with access call box.*
*Overhead downlights at soffit above.*

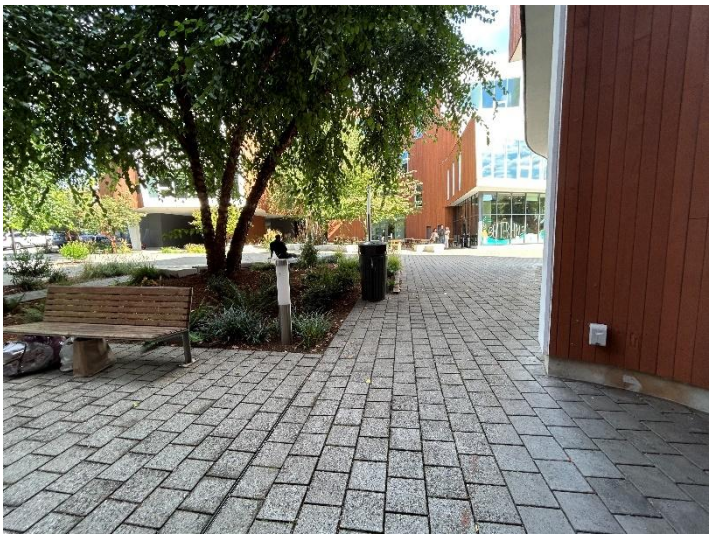


*Figure A-14 Karuna II West courtyard view from exit passage to fire lane beyond. Common courtyard fixtures include lighting, waste receptacles, and benches.*

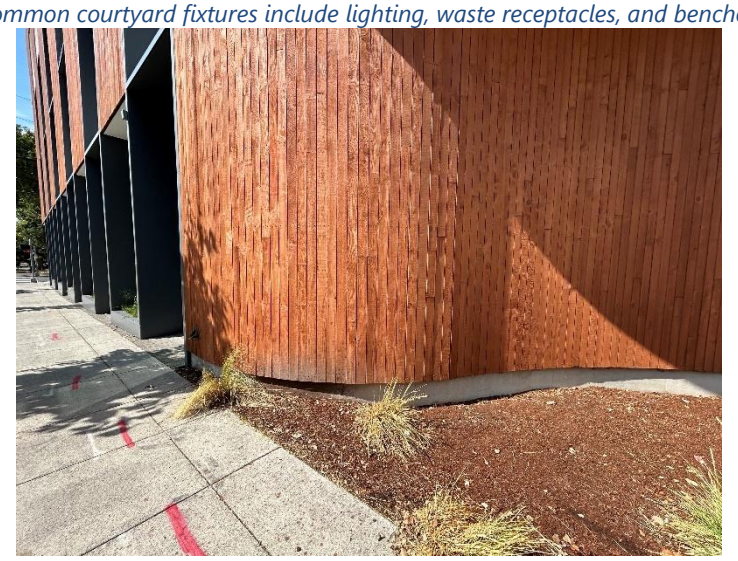


*Figure A-15 Karuna II West southeast building corner at Fremont. Base of wood siding less than 7 inches above ground level. Debris on wood veneer. Grass plantings are fair to poor condition.*

**Recommendations**

- Measure exit illumination levels along path from exit to fire lane.
- Maintenance to remove discoloration on concrete stem wall.
- Maintenance to secure paver system smooth and well anchored. Paver joint with metal insert extends to building from planter.
- Paver wear varies at building wall and beyond drip line.
- Existing exterior furnishings and fixtures—including exterior lighting, waste receptacles, and benches in the courtyard—present safety and fire risks. Recommend evaluating these fixtures are

securely anchored to prevent vandalism or displacement, and are strategically placed to support security, sanitation, and fire prevention protocols.
- Repair damage exterior metal panels Karuna II East covered parking walls and install barriers to limit future damage from cars.
- Our observation was during daylight hours. Recommend PPS Risk visit the site during night hours to assess lighting levels in courtyard and parking areas for security and safety.
- Due to "woonerf pedestrian street" design of alley with adjacent courtyards, no striping or other obvious indication exists to delineate vehicular vs pedestrian use. Recommend that PPS Risk evaluate courtyard conditions to determine if bollards or other means to delineate this division of traffic should be installed for safety of occupants.

## Building Exterior Conditions

Completed in 2015, One North set a regional precedent for commercial office buildings in Portland through its distinctive architectural design and material palette. The development features exposed wood structural framing, operable windows, expansive upper-level terraces, and open, flexible interior layouts—elements that reflect the CBSE program's emphasis on innovation, sustainability, and adaptability. Generous natural daylight and views of both the city and surrounding landscape further enhance the building's creative and collaborative environment.

The exterior design remains uncommon for 4- and 5-story commercial buildings in Portland. Curved walls clad in stained wood veneer, white-painted projecting apertures, and low-profile roof coping and flashings contribute to the building's unique character but require more frequent upkeep than the PPS District Standard brick and metal panel systems, which follow a 20-year maintenance cycle.

Since its construction, the building has been maintained by reputable firms including [redacted] These companies have provided ongoing service for roof systems, mechanical and electrical infrastructure, and fire protection systems. In 2024, exterior maintenance included repainting the apertures and refinishing the wood-clad curved walls, preserving the building's architectural integrity and performance.

**Building exterior doors swing out to the sidewalk and courtyard exit direction.** Entry/exit doors to lobby and other street level spaces do not have standard PPS door stop bollards. If door-stop bollards are desired, it may be unlikely that they can be installed in all locations, due to the cost of demolishing the sidewalk or pavers and the location on street sidewalk. Incorporating card swipes and ADA buttons while limiting surface-mounted elements requires further study.

**Building lobby and tenant spaces do not have exterior air lock vestibules.** No vestibules are provided at the building entry lobby, entry from site is directly into the lobby. Walk off floor surface is provided. Providing exterior vestibules to reduce energy loss and provide secure entry area is common for school buildings.

The ownership of commercial office buildings with tenant agreements comes with building management and maintenance that vary from school buildings. Operations costs with regularly scheduled maintenance and repairs will preserve the development character and set precedent for the program.

A starting point for planning purposes:

| Exterior Materials | Maintenance | Replacement/ Major repair | Project Year from 2024 |
|---|---|---|---|
| Stained wood veneer | Annually | 3-5 years | 2027 |
| Joint sealants | Annually | 3-5 years | 2027 |
| Aperture | Annually | 3-5 years | 2027 |
| Steel Frame Planter and Shade | Annually | 15-20 years | 2027 |
| Exterior doors and frames | Annually | 15-20 years | 2027 |
| Window system | Annually | 15-20 years | 2030 |
| Operable windows/ screens | Annually | With Windows | 2030 |
| Roof drains | 2 times per year | 20 years | 2035 |
| Terrace Pavers (roof) | 2 times per year | 20 years | 2035 |
| Building roof | 2 times per year | 20 years | 2035 |
| Solar panels | Monthly/ Quarterly | 15-20 years | 2030 - 2035 |

#### Recommendations

- Obtain the building and site maintenance and repair records for PPS to reference.
- Transfer building material and manufacturer warranties to new Owner.
- Develop O and M schedule and budget to continue consistent and proactive exterior maintenance for the wood veneer, sealants, drains, and roof flashings.
- Evaluating the building entries for energy and building code conformance is recommended for energy savings and potential security measures.

## Apertures, Terraces, and Roofs

Both the Karuna II West and East building exterior walls curve and offset from floor to floor creating uniquely shaped overhangs and roof areas. Apertures frame windows, and the shifts in floor plate provide overhangs, roof areas with terraces, and roof areas above. Building roofs slope to interior roof drains. Drains collect in the ceiling space below and travel towards central area before dropping vertically to storm line below.



*Figure A-16 Karuna II roof to exterior wall and roof drain.*
*Install roof drain screen and anchor into flange.*

Terraces extend the views and usable space for upper tenant level spaces. Exterior swing doors from tenant spaces walk out onto concrete pavers. The terrace pavers with open joints conceal primary and overflow roof drains on the roof surface below. Glass guard rails maximize the attractive city views and with their position close to the edge of the building and "on grade" shelf on the opposite side present a concern for use by students or public if someone climbed it.

Once on the roof terrace – access back into the building does not appear to be controlled. Based on prior PPS projects with roof terraces, PPS typically would want to control access after-hours and may require a call box. Retrofit work and an appeal may be necessary to get access control and hardware working in accordance with PPS typical operations, for PPS Risk review.

Roof finishes are white with roof membranes adhered to low profile coping. Dirt and debris on the white membranes are visible through tenant windows from the interior and adjacent buildings. Frequent maintenance will be necessary to keep the area clean and avoid deterioration. The only way to clean these areas would be via vertical lift from the street and courtyard.




*Figure A-17 Karuna II East building white apertures surrounding upper-level windows.*

The building's primary roof is access by ships ladder through large roof hatch. The ships ladder and hatch provide ease of access from stair below. The hatch lid when open on Karuna II East building provides a barrier at the north building roof edge. A short section of raise parapet provides a tall guard when stepping onto the roof. The roof hatch on Karuna II West building is centrally located within a few steps of the main mechanical equipment enclosure.

Rooftop features include a mechanical screen enclosure, raised roof over the elevator shaft, and large area of solar roof panel racks. Solar racks rest on top of the roof membrane and ballasted with concrete blocks. Walkway widths for maintenance of solar roof panels were less than required along the roof perimeter to panels do not conform with code.

Main building roof has various heights with the tallest transition to elevator roof measure approximately 22 inches from the main roof. Conditions at the roof perimeter vary with low raised parapets and stepped roofs. Traditional raised parapet with coping cap varies in height from 9-14 inches at the roof perimeter. Step along perimeter of roof varies in height from 6-12 inches before extending out to finish with a low slope coping beyond.

Painted steel mechanical screens surround roof top mechanical equipment near the roof centers. Roof top equipment on raised curbs, surface conduits, and pipes on blocks, and control panels mounted to the screen frame can be found within the enclosure. Doors on the enclosure are heavy steel with lever hardware and latches. The doors have sagged over time and require effort for them to close.

Fall protection exists with a single tie off near the roof center and additional D-rings welded to the screen tube steel posts of the mechanical screen enclosure. Exterior lighting was not observed on the roof level.

**Recommendations**

- Secure roof drain screen baskets to trim ring.

- Adjust mechanical screen enclosure gates to remove sag and result in ability to open and close without force.
- Adjust solar panel racks to provide walkways at roof perimeter to conform with code.
- Evaluate installation of additional fall arrest anchors for worker safety tie off.
- Engage PPS Risk to evaluate roof terraces for student and public use areas. The guard rail to roof ledge configurations and access control after hours.



*Figure A-18 Karuna II West unique roof shapes.*
*View north from tenant space level 4.*



*Figure A-19 Karuna II West. Roof drain at exterior wall.*
*View east from level 4 tenant space.*



*Figure A-20 Karuna II West. Roof drain at exterior wall.*
*View west from level 4 tenant space.*



*Figure A-21 Karuna II West South terrace doors in alcove with sprinkler and downlight.*
*Downlight repair recommended.*
*South terrace at level 2.*



*Figure A-22 Karuna II West South terrace door alcove from tenant space.*
*Verify terrace door opening forces conform with accessible requirements.*

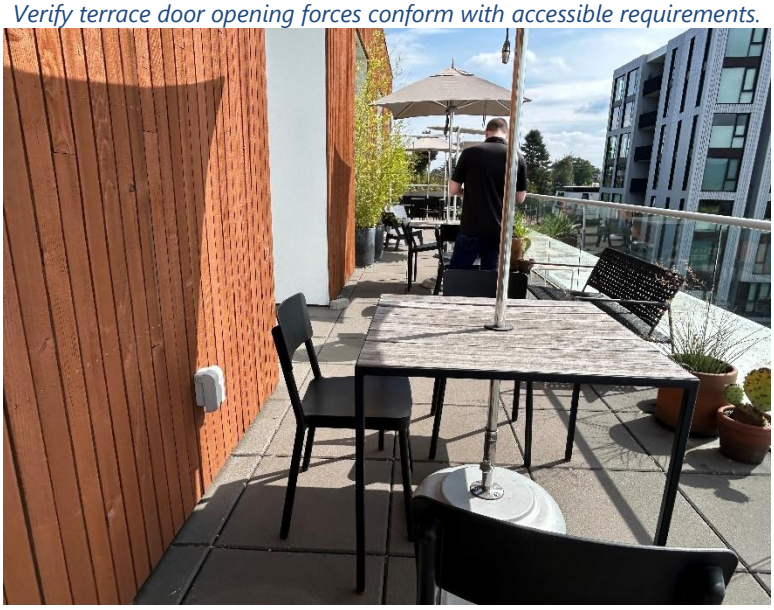


*Figure A-23 Karuna II West south terrace view east.*
*Concrete paver finish over roof with drains below.*
*Glass guard rail height at perimeter conforms with code minimum.*



*Figure A-24 Figure 48 Karuna II West Roof Access hatch.*



*Figure A-25 Karuna II East Roof Access hatch and perimeter guard at north building wall.*



*Figure A-26 Karuna II West 'D ring' anchor on mechanical screen enclosure post.*



*Figure A-27 Karuna II West fall restrain anchor at main building roof.*



*Figure A-28 Karuna II West view inside main building roof top mechanical screen enclosure.*

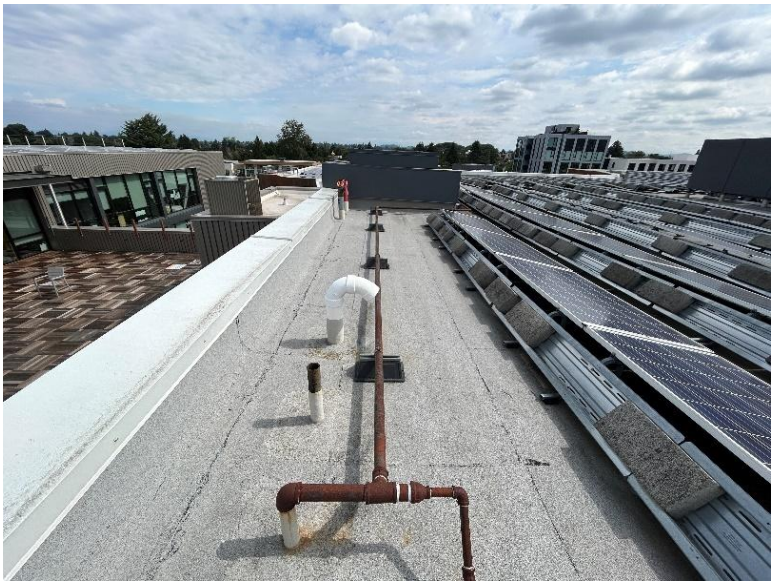


*Figure A-29 Karuna II West main building roof parapet coping at north wall.*
*Gas line on surface blocks.*
*Standpipe and elevator roof view beyond.*



*Figure A-30 Karuna II West building roof step at south wall.*



*Figure A-31 Karuna II West building roof step at northeast corner.*



*Figure A-32 Karuna II West roof at elevator roof and north property line.*
*Metal joint cover transitions from building roof to elevator roof clad in metal roof panel.*
*Fire standpipe onto roof from stair below.*

## Building Interior

For both Karuna II West and East buildings the overall building interior aesthetic is open, warm, and inviting with exposed concrete and mass timber on floors 2 and above. Mass timber is new material for Portland Public Schools but has been used on at least one recent project. Concern over damage to the material by denting or marking due to user or maintenance interference does not appear to be an issue in practice.

## Common Shared Elements

The Karuna II West and East buildings each have main building entries with public street facing doors and main lobbies connecting through to the courtyard. Street level tenant spaces have street or courtyard facing exterior doors. The main lobby for each building provides access to fire sprinkler room, electrical room, elevator and elevator room, and waste collection room with convenient street level access. The lobby and waste collection room is a common, shared element between building tenants. Shared spaces at main level include gendered toilet rooms and separate locker room with shower facilities.

Upper levels of the buildings vary with tenant space layouts.

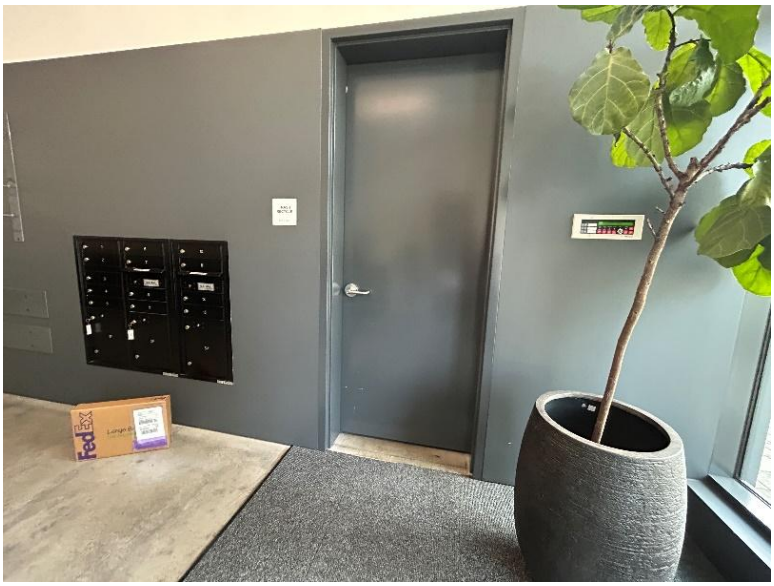


*Figure A-33 Karuna II West Street entry lobby.*
*Mail and Waste collection room access point for building tenants.*

## Exit Stairs

Each building has two interior exit stairs and an elevator. The elevators open onto landings at each floor above street level. Elevator smoke doors and enclosed lobbies were not observed during the site visit.

The exit-stair enclosures are rated walls and doors fitted with closers and latching, lever hardware. Stairwells are finished with painted gypsum board walls and rubber base. Stair construction is wood with rubber nosing, treads, and stringer face. Stair handrails are wall-mounted painted steel. A center wall between stringers provides continuous wall separating stringers. Guard walls at the top floor landings are finished with painted gypsum board measuring more than minimum 42 inches above finish floor in compliance with applicable codes.

One stair for each building includes fire riser and roof access by way of ships ladder at its highest floor level. Roof hatch size conforms with code and district standards provided convenient access for maintenance work. Stair signage within the stair conforms with a code.

Each stair was visually reviewed for consistency of finish and condition. A random sampling of stair rails, riser, tread, and walkline measurements found riser height, tread depth, handrail height, continuity, and termination, landing depth and walkline to be in close conformance with applicable codes.




*Figure A-34 Karuna II West and East Stair signage conforms with code.*

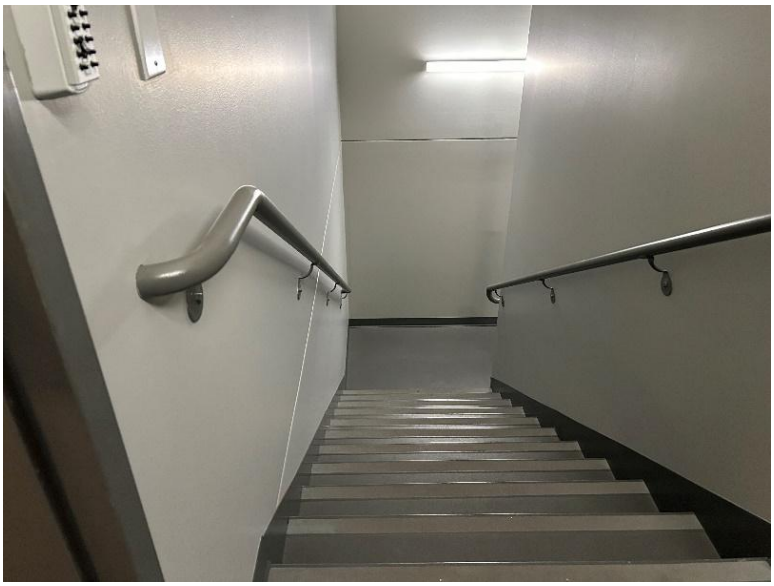


*Figure A-35 Karuna II East and West Interior Exit Stair typical finishes*



*Figure A-36 Interior Stair guard wall and ship ladder roof access at top landing.*

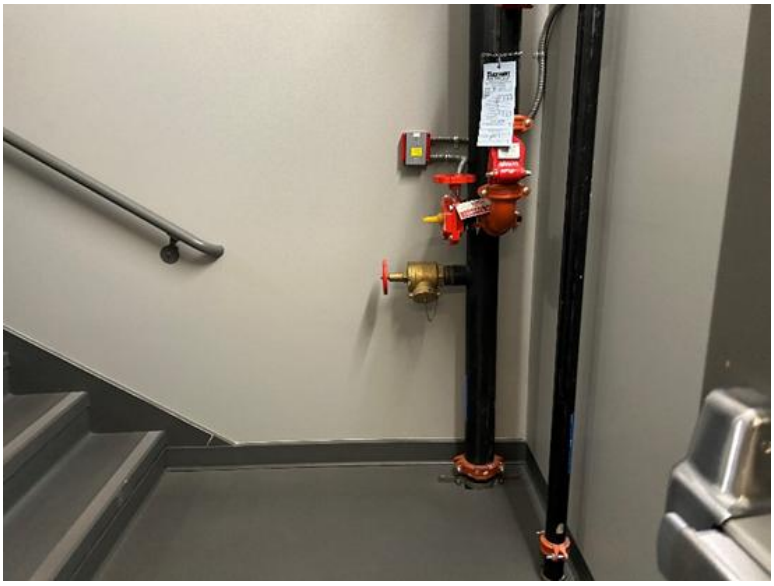


*Figure A-37 Karuna II West and East stair landing walk lines.*
*Fire standpipe drain valve beyond walk line conforms with codes.*

### Recommendations

- Karuna II West Stair 2: Minor maintenance is needed to secure loose rubber stringer base near mid-level landing Level 2 and below in Karuna West Stair 2 (south stair).
- Level 1 Karuna West Stair 2 Level 1: This stair exits to the Courtyard by exit passage. The landing depth at base of stair is tight fitting space compounded with the inside rail termination projecting beyond the wall. The rail termination does not conform with code. The return of the rail to the wall was a commonly accepted practice at the time of construction and presents a hardship for modification to conform today. Recommend review of building code appeals submitted at time of construction.

## Toilet Rooms

Karuna II West and East buildings each provide common use gendered toilet facilities including main level locker rooms with shower facilities. Restrooms are gendered, which deviates from the all-user approach taken with the most recent PPS school modernizations and new construction. A janitor's closet with mop sink is located near the facilities. Room signage for these spaces is accessible with exception of the Karuna II East Levels 2-5 and within Karuna II West Level 3 full floor tenant space.

Karuna II East building toilet facilities are unique to each floor and tenant. The street level healthcare tenant's space has two single-user rooms to support patient functions. The healthcare and café tenant spaces share gendered toilet rooms with two compartments and two lavatories. Employees and café patrons have access to this common hallway between tenant spaces. This hallway also provides a second exit for the café tenant that also leads to an exterior exit door.

On floors 2-4 in Karuna II East, the toilet rooms are located near the elevator, but not in view, and efficiently stack from floor to floor. Room signage on these floors is unique and not consistent with accessibility standards. Fixtures within the room match the development standards with floor-mounted partitions, toilets, and wall hung urinals. Accessory finishes vary from building standard finishes in this tenant space.

Karuna II West building toilet rooms stack from floor 1 to 5 and share a common layout at each level. Floor Level 1 has two lavatories in each toilet room unlike floors 2-5 above with a single lavatory in each. Floor finishes are polished concrete or resilient flooring with rubber base.

Building standard finish for fixtures is white with polished chrome. Fixtures are floor mounted with exception that urinals and lavatories are wall mounted. Faucets are typically hands-free Kohler brand and vary in style by tenant. Gendered rooms with multiple fixtures are fitted with floor mounted metal partitions with finishes that vary by floor level and tenant. Accessories including grab bars, toilet paper dispensers, toilet paper dispensers, toilet seat covers, and sanitary napkin dispensers are satin stainless steel and brushed chrome typically. Paper towel and soap dispensers vary in location, brand, and mounting. The East building tenant space on floors 2-4 has a custom color finish for accessories.

Primarily walls are finished painted gypsum board with rubber base except wet areas at showers and mop sinks where wall protection is installed. From a maintenance perspective, the toilet rooms are not provided with tile or other durable wainscot materials common in PPS facilities. Tile wall finish was observed in the first floor changing and locker rooms. Ceiling finishes vary from metal panel in a suspended grid to painted gypsum board and are most commonly open to structure above.

Toilet room doors are equipped with closers. Single user rooms have thumb-turn latch in addition to closer ones. Shower and changing room doors have secure access by card reader or keypad. Flooring finishes are concrete or solid surface with slight slope to a common floor drain.

Overall toilet rooms are well maintained and accessible with few exceptions.

### Recommendations

- Mounting of accessories below grab bar interferes with grab bar operations and is recommended to be relocated per ADA standards.
- Wall-mounted paper towel dispensers on site measured more than four inches and where they are located along the accessible route from toilet room door to accessible stall are considered protruding objects. Installing a maximum 4-inch-deep dispenser is recommended at these locations.



*Figure A-38 Karuna II West Street Level gendered toilet room finishes.*

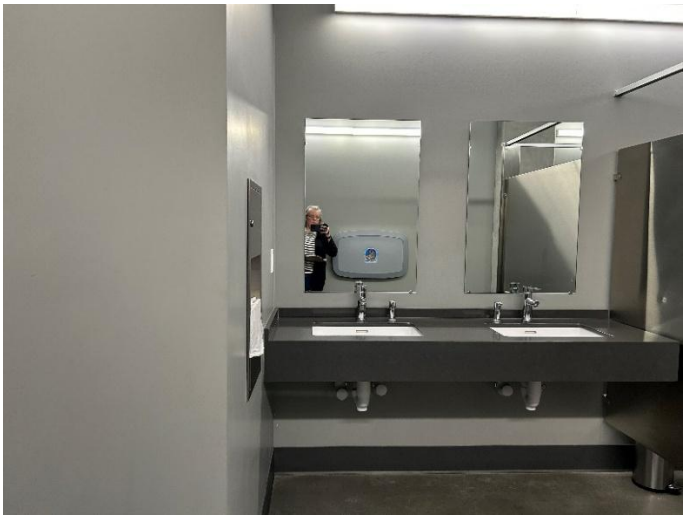


*Figure A-39 Karuna II West Street level gendered toilet room finishes.*



*Figure A-40 Karuna II West Upper-level gendered toilet room finishes.*



*Figure A-41 Surface mounted Paper towel dispensers protrude more than four inches from wall on the route to accessible toilet compartment.*



*Figure A-42 Wall hung lavatory maneuvering clearance and hands-free operating faucet provided.*



*Figure A-43 Typical accessible toilet and grab bar installation.*



*Figure A-44 Grab bar and Toilet Accessories surface mounted.*
*Projection and location of accessories conflict with accessible clearances for grab bar.*

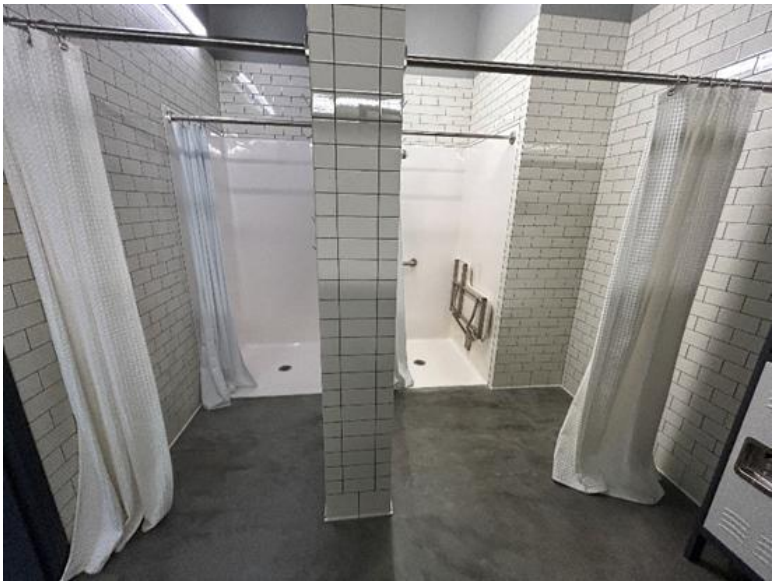


*Figure A-45 Main Level Changing room showers*

## Elevators and Landings

Both Karuna II West and East buildings have a single-cab Schindler elevator connecting the floors. Access for visitors is elevator-only. Elevator cab finishes are stainless steel with handrails mounted at sides and rear cab. Elevator call buttons at each floor are located at the hoistway door side jamb within reach range for accessible maneuvering clearances. Elevator hoistway doors are stainless steel finish, similar in appearance to the building common area hardware finishes.

Elevator landings on Level 2 and above open directly onto the floor. In 4- and 5-story buildings equipped with automatic fire sprinklers, enclosed elevator lobbies are required except by building code exceptions where additional smoke doors or curtains are provided. Smoke containment at elevators was not reviewed for code conformance during the site visits. Signage was minimal at the landings, and no signage was observed that indicated elevators should not be used for exiting the building.

Karuna II West building has four stories above street level and requires an elevator equipped with standby power for accessible egress. A roof mounted natural gas generator serves the elevator emergency power for this building only. This calls attention to the need for building signage marking accessible route across site, within buildings, and exiting.

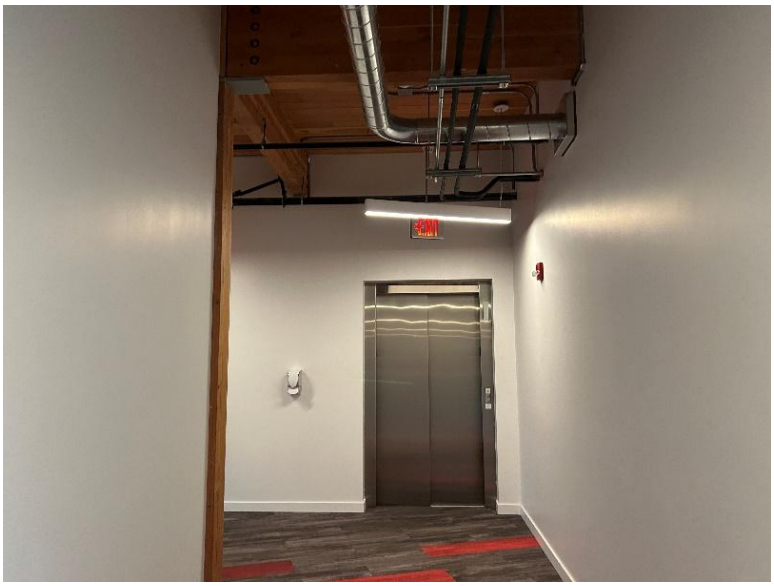


*Figure A-46 Karuna II West Elevator Landing Level 2*



*Figure A-47 Karuna II West and East building elevator cab interior*



*Figure A-48 Elevator call buttons mounted on frame.*

## Tenant Spaces

Existing Karuna II West and East buildings vary in the number and size of tenants. Both the West and East building street levels currently provide space for two tenants in addition to the main lobby, service rooms, locker rooms, and long-term bicycle parking.

### Floor Plans and Finishes

The East building has a larger "U-shaped" floor footprint than the West buildings "L-shaped" plan. Both buildings provide some common elements (elevator, two interior exit stairs, and gendered toilet rooms) on each level. The West building's smaller floor footprint was constructed with the flexibility to have a single or multiple tenants on each floor and features a 5$^{th}$ floor making the building similar in building area.

The building interior finishes and fixtures observed on site are industry standard 'vanilla box' for commercial tenant spaces, providing a clean base for tenants to customize for an additional cost. The First Floor Street level tenant spaces feature concrete floors, aluminum frame windows and doors, exposed structural concrete columns, and structural slab ceiling above.

Upper-level floor finishes included variety of exposed concrete, carpet tile, resilient sheet flooring with wall base finishes of rubber and wood. Within the tenant spaces, wall finishes were customized with writable surface paint or magnetic metal surfaces, wood cladding and wallcoverings. Common hallway areas were observed to be gypsum board without a wainscot material or plaster finish necessary to withstand the heavy impacts common in schools and student use areas.

The East building, floors 2 and above, was built as a single multi-level tenant space with a central atrium and open stair. This unique space features tenant-specific finishes like stadium seating, an open stair, metal guards lining the atrium level openings, and fire shutters at the top floor to separate it from the floors below. The toilet room fixture layout, location near elevator landing, and accessories vary from the West building. This open atrium area with its handsome learning stair constructed of durable materials would be beneficial for gatherings and presentations and comes finished with existing motorized shades.

The open stair connects levels 2, 3, and 4 within the tenant space. The open stair guard is made of black steel and welded wire mesh panels that appear sturdy and easy to maintain. The perimeter balcony guard is provided with built-in tables for use as a "brain bar" or ledge to sit at with stools. Design detailing and install of the stair handrail leaves some hard, sharp edges and corners that may be of concern in an environment used heavily by students. The ledges provide useful work and social space but may be of concern due to location adjacent to "open to below" space, and any objects placed on these ledges (i.e., potted plants) could be easily pushed off and over the edge of the balcony.

Conference/office spaces are provided along the west side of the East building tenant space on all levels 2-4. The floor-to-ceiling black steel and glass walls are attractive and sturdy but may be of concern due to their minimalist design since it would be easy to walk into the glass. Due to the black steel frames and heavy glass, the doors to these rooms are very heavy and the pound force needed to operate them appears to exceed code. Door buttons have been provided in these rooms to "request exit."

Polished concrete and carpet tile flooring in the East building suite is suitable and in keeping with PPS standards, however the base detail utilizing a small painted wood shape at the intersection of wall to floor will be difficult to maintain per standard PPS cleaning practices. Carpet installation where floor meets the wall was inconsistently cut and the base did not lap the carpet in all areas.

### Kitchens and Utilities

Tenant spaces in both buildings include kitchen spaces varying from the street level restaurant with commercial equipment and walk-in freezer to office kitchenettes. The interior fit out of tenant spaces is designed by tenant in agreement with building management and as approved through building permits. This custom approach allows variation for tenant satisfaction and adds complexity to building operations and maintenance.

Several upper-level tenant spaces in both buildings are built out with linear style kitchen spaces and 34-inch-high counter tops. Casework varies in finish as selected by the tenant. The kitchen spaces do not stack from floor to floor in the buildings. The utilities for power, drain, vent, and gas lines have paths leading back to common walls near stairs and elevators, typically. These lines are exposed to ceilings on the floors below and in some cases require special access panels and hatches where tenants have finished ceilings. Electrical panels were not all located inside rooms. Panels located in common areas were in flush-mounted cabinets with locks.

Fixtures include sink, faucet, dishwasher, and appliances, which vary with refrigerators, microwaves, and, in one West building kitchen, an electric range with hood. The East building primary tenant space features a commercial gas range with hood on its top floor level. Additionally, the East building top level features a gas fireplace appliance, unique to office tenant space.



*Figure A-49 Karuna II - Tenant Shell Space - Street Level*



*Figure A-50 Karuna II West Level 1 Storefront sill at curb condition*

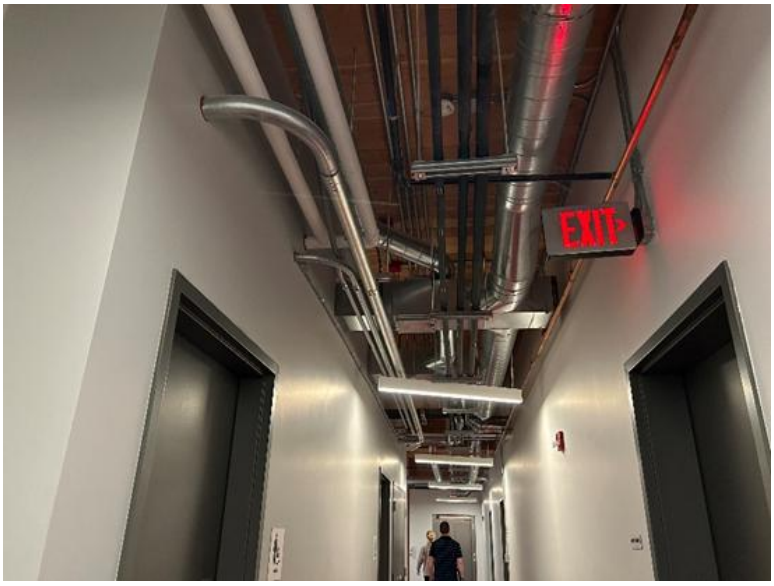



*Figure A-51 Karuna II West Level 2 - Corridor for multi-tenant access.*

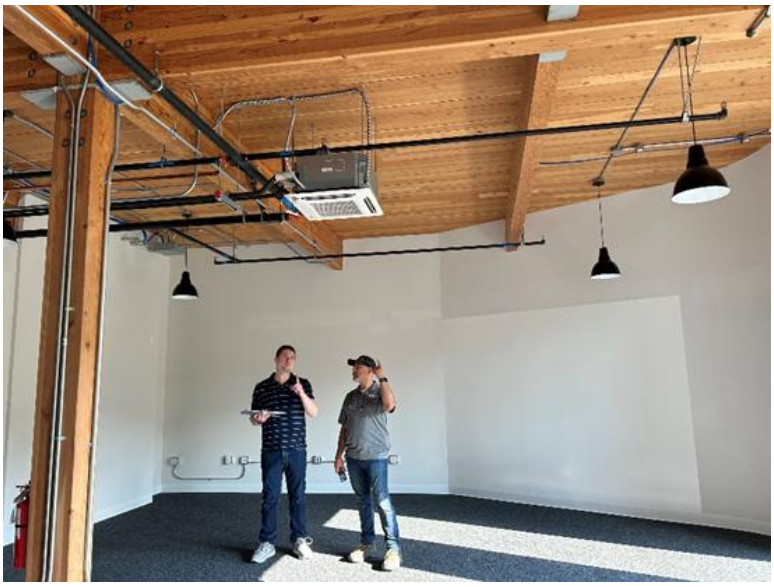


*Figure A-52 Karuna II West - Tenant Shell Space.*

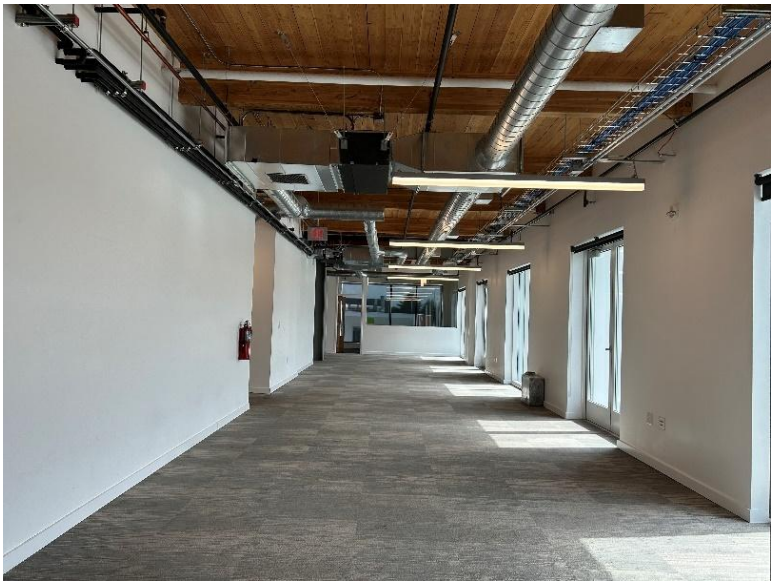


*Figure A-53 Karuna II West - Tenant shell space.*

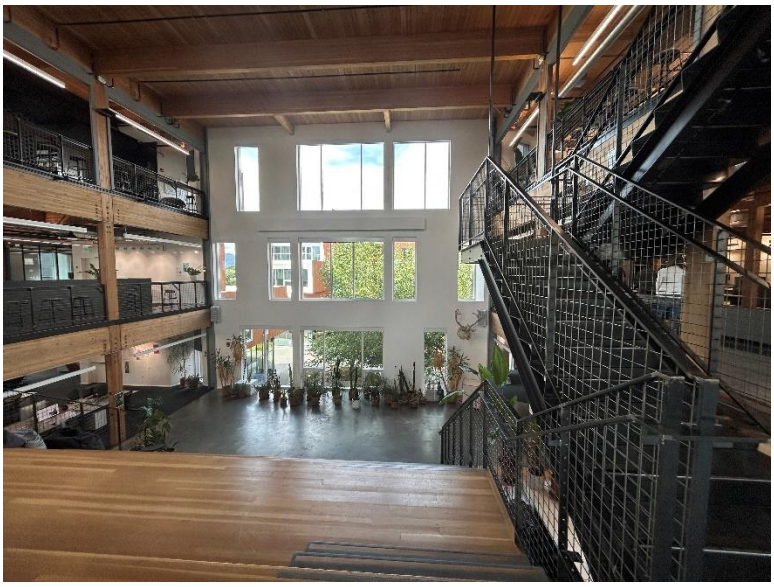


*Figure A-54 Karuna II East stadium seating and open stair in tenant space.*
*Open space connecting 3 floors with fire shutters at level top level.*
*Steel stair with guard and continues rail.*
*View west to courtyard beyond.*

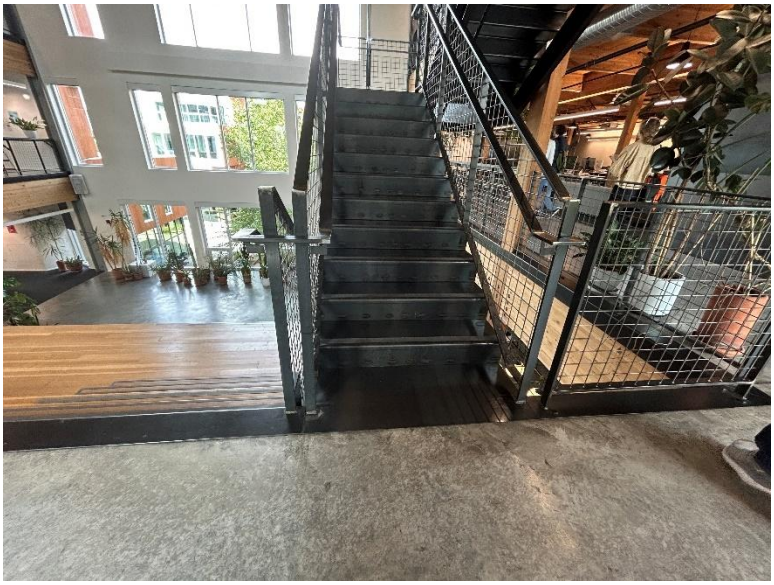


*Figure A-55 Karuna II East open stairs at tenant space Level 3.*

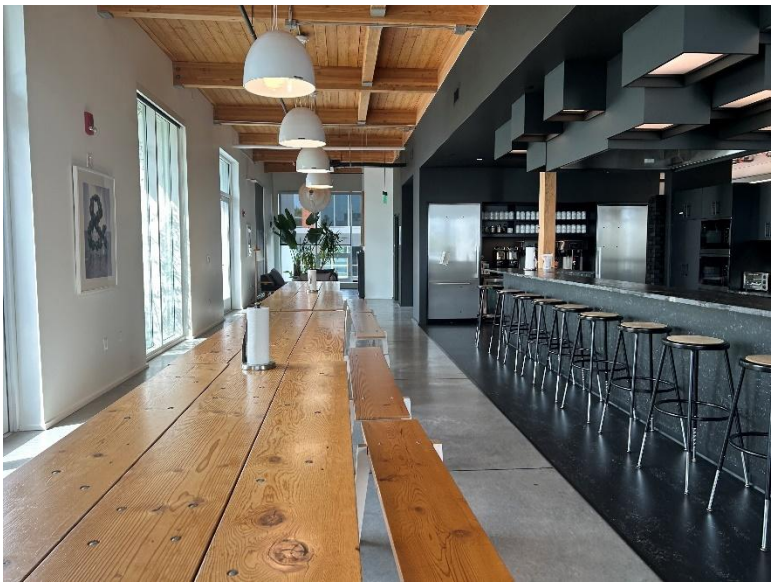


*Figure A-56 Karuna II East tenant space Level 3 kitchen with terrace doors to left.*



*Figure A-57 Karuna II East tenant kitchen gas range with hood above.*

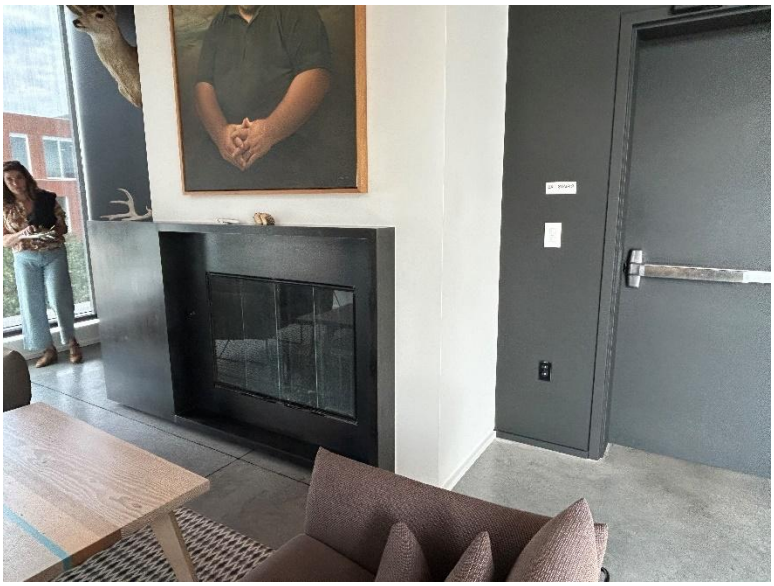


*Figure A-58 Karuna II East tenant space with wood fireplace and exhaust through roof above.*



*Figure A-59 Karuna II West - Tenant space kitchen*

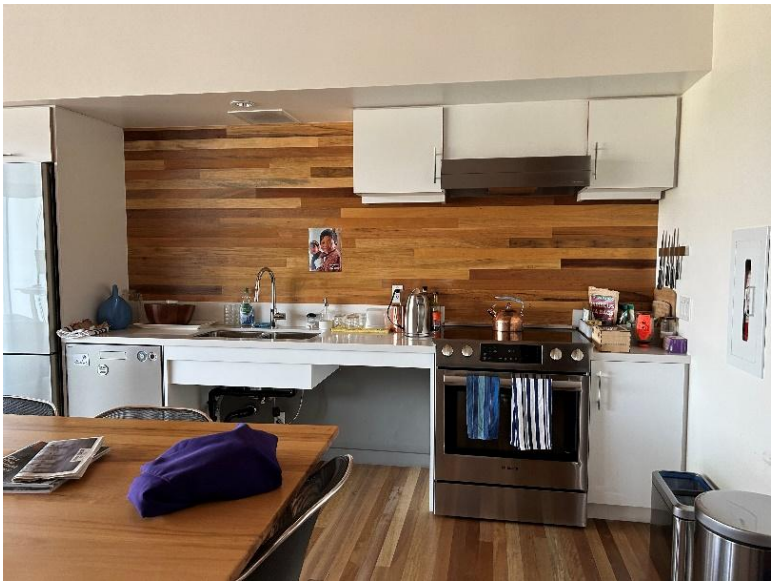


*Figure A-60 Karuna II West - Tenant kitchen with electric range and hood.*

## Building Signage

The building signage for Karuna II West and East buildings is similar in theme and style for the two buildings. Exterior building signage at the main street entries is applied to the storefront glass. Tenant spaces at street level also have signage near their street entry doors. Size and color of tenant signage varies as allowed with city permit standards.

Service rooms, common use rooms have signs indicating the use. Sign types vary and are typically raised letters in contrasting color to the background. Braille below the raised letters is typical at common area signs including the tenant side of exit stairs.

Signage at elevator landings is minimal. Accessible route and exit access signage inside and outside buildings was not observed to connect primary elements. Code required accessible exit signage was not observed in either building.

Typical signage for toilet rooms, interior exit stairs  conforms with code for mounting height, symbols, raised characters and braille. Conformance varies inside tenant spaces.




*Figure A-61 Karuna II West - Typical gendered toilet room signage*



*Figure A-62 Karuna II West - Tenant space toilet room signage. Lettering applied to wall surface is not raised and does not conform to code for accessible toilet room wayfinding.*

## Doors and Hardware

Door and frame materials vary for the buildings. Exterior doors and frame types include aluminum storefront or hollow metal with insulated steel. Storefront doors have narrow frames and low aprons where the steel doors are single panel without windows. Service room and exit stairs are fitted with single panel, painted steel doors and frames.

Tenant entry doors are general stained wood finish with frosted glass full light. Variations for painted finish on tenant entry were observed on site in the Karuna II West building.

Door hardware is brushed chrome finish across the site except unique conditions within the Karuna II East building tenant space levels 2 and above; the exterior concealed hinges; and mechanical equipment screen gates. Lever action hardware exists with varying profiles. Exit doors were observed to have closers with latches and panic bars.

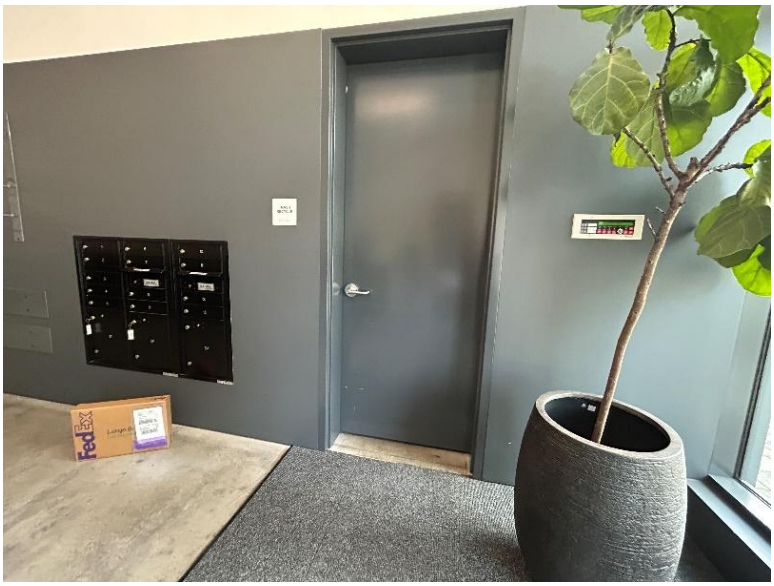



*Figure A-63 Karuna II West - Main building lobby shared use mail and trash collection access.*




*Figure A-64 Karuna II West and East buildings – Service room door and signage – Main lobby*



*Figure A-65 Karuna II West - Tenant entry door*



*Figure A-66 Karuna II West - Tenant entry door wood glazed door and hardware*



*Figure A-67 Karuna II West - Tenant entry door wood glazed door and hardware*

## Windows and Treatments

Karuna II West and East buildings the exterior windows are aluminum with insulated glass. Street level aluminum storefront window and doors are dark colored. The street level window sills are anchored on low concrete curbs. At upper-levels the exterior windows and doors are white aluminum frame and on terraces the sills are anchored at the floor. Generally, all windows are fixed lites with their Interior window opening surrounds finished in painted gypsum board. Operable windows, integrated into the window systems on upper floors, have screens and limiters that prevent the window opening more than four inches. The operable windows provide tenants with ability to control fresh air flow in conformance with LEED building standards.

Windows treatments in tenant spaces are consistent in color and style through both buildings. Window treatments provide building users the ability to control the amount of natural window light and adjust privacy from exterior views. Shades are roller style with variation of manual and motorized operation. Black out shades in addition to perforated light shades occur in some spaces. Shade style is consistent with PPS standards.

The East building primary tenant space features conference rooms and office spaces on the west side of levels 2-4. The floor-to-ceiling black steel and glass walls separate these spaces from the open office. The Glass walls are attractive and sturdy but may be of concern due to their minimalist design since it would be easy to walk into the glass. Due to the black steel frames and heavy glass, the doors to these rooms are very heavy and the opening force needed to operate them appears to exceed code. Door buttons to “request exit” are provided in these rooms.

In the West building the full floor tenant Level 3, private offices and other rooms are fitted with access control swipe pads to secure the rooms. The suite also has motorized black out shades with switch controls near the room entry doors.



*Figure A-68 Karuna II West and East building roller shade window treatments*



*Figure A-69 Karuna II West and East building windowsill finish*

### Recommendations

- Recommend consideration of more durable material such as solid surface at windowsills in areas of student use areas.
- Recommend evaluation and consideration to install durable, heavy impact finishes in areas where student use is planned.
- In East building primary tenant space floors 2-4, recommend that PPS Risk review conditions:
    - Around learning stair, recommend at a minimum that “brain bars” be removed or that decorative panels be added that provide a barrier.
    - Conference/office spaces glass walls and doors to determine if glazing film is desired, or if the renovation replaces these with more conventional storefront/doors and hardware.
    - Roof terrace door access control.
    - Roof terrace guard rail to roof ledge configurations.
    - Review the fireplace with glass doors and a black steel surround installed on the 4th floor, south for consideration to remove the appliance. The layout of this fireplace and space behind it creates a “hidden area” west of the fireplace, with poor sightlines.
- Where electrical and service panels were not all located inside locked rooms. Recommend securing the panels with locks where visible and accessible to view in common areas.
- For O and M purposes, recommend an evaluation locations where motorized and manual shades occur.

**END OF ARCHITECTURAL SECTION**

# Structural

## Summary of recommended but not required Structural cost items

- As with many mass timber structures, some shrinkage has occurred over time, resulting in localized gaps at connections where bolts or screws have loosened. This shrinkage should be substantially complete at this stage, and it is recommended that any loose bolts and lag screws, exposed to view be tightened to maintain proper bearing at connections.
- Overall, no conditions were observed that would indicate a reduction in the structural integrity of the building.

## Discussion West Building "Karuna II West"

Karuna II West is a five-story office building constructed in 2014 under the 2010 Oregon Structural Specialty Code. The structure was designed as a Risk Category II facility, consistent with typical mid-rise office occupancies. The first story is composed of reinforced concrete construction, with a 12" to 14" thick post-tensioned concrete slab at the second level supported by concrete columns and concrete shear walls. Above the concrete podium, the building transitions to mass timber framing. Floor construction at the upper levels consists of 5/8-inch plywood over 4-inch Lock-Deck panels, supported by glulam beams and columns. Lateral resistance in the mass timber portion is provided by plywood-sheathed shear walls, supplemented by a pair of steel moment frames. The foundation system consists of reinforced concrete spread footings bearing on engineered granular pads. The roofs support a ballasted PV array as well as mechanical equipment that is located behind a screen wall.

Karuna II West has been in service for approximately eight years and, as a relatively new structure, appears to be performing as expected. The mass timber framing did not exhibit significant checking or splitting; while some minor checking was observed, it is consistent with the natural behavior of timber and considered typical for this type of construction. Minor drywall cracking was noted in isolated locations, particularly at re-entrant corners, which is also common in buildings of this age and structural system. A few timber connections exhibited visible gaps due to shrinkage over time.

In reviewing the structural drawings, we noted that the post-tensioned slab at Level 2 was not detailed to accommodate shrinkage in the manner typically expected. Standard practice often includes a pour joint near the mid-span of the slab or a slip joint at the slab-to-wall interface, which allows the slab to shrink before a pourback is placed. Without this detailing, slab shrinkage can impart additional stress on the concrete shear walls as they resist the slab's shortening. The shear walls were not exposed to view during our site visit, and no shrinkage-related cracking or distress was observed in the accessible areas. This is a condition to be aware of if cracks are discovered hidden under finishes.

# Discussion East Building "Karuna II East"

Karuna II East is a four-story office building also constructed in 2014 under the 2010 Oregon Structural Specialty Code. Like its counterpart, the building was designed as a Risk Category II structure. The first story utilizes reinforced concrete construction, with a post-tensioned concrete slab at the second level supported by concrete columns and concrete shear walls. The upper three stories are framed in mass timber, with typical floor construction of 5/8-inch plywood over 4-inch Lock-Deck panels supported by glulam beams and columns. Lateral resistance above the podium level is provided by plywood-sheathed shear walls and a pair of steel moment frames. Foundations consist of reinforced concrete spread footings bearing on engineered granular pads. The roofs support a ballasted PV array as well as mechanical equipment that is located behind a screen wall.

Karuna II East has also been in service for approximately eight years and is generally in good condition for a building of its age and construction type. The mass timber elements showed only minor checking and splitting, which is typical and not considered structurally significant. Some water staining was observed on glulam columns and beams, most notably near column-to-floor joints. Based on the staining patterns, this appears to result from routine floor cleaning, where water has seeped through small joints at the floor/column interface and run down the face of the columns. The staining is considered an aesthetic issue and does not affect the structural performance of the timber. Similar to the West building, small drywall cracks were observed in locations with re-entrant corners, and a few timber connections exhibited visible gaps due to shrinkage over time.

As with Karuna II West, our review of the structural drawings indicated that the post-tensioned slab at Level 2 was not detailed with a shrinkage joint or slip joint at the slab-to-wall connection. This type of detailing is often used to reduce stresses on shear walls by allowing slab shrinkage to occur prior to placement of a pourback. In the absence of this detail, shrinkage of the slab can increase stresses at the wall/slab interface. During our observations, the concrete shear walls were not exposed, and no visible shrinkage-related cracking or distress was noted. This condition should be monitored in the future, particularly if cracking develops at shear walls or at slab-to-wall connections.

# Recommendations

**Code Implications of Occupancy Change**

If the Karuna II buildings were converted from office use (Group B, Risk Category II) to school use (Group E), the Oregon Structural Specialty Code (OSSC, based on the 2021 International Building Code) would require evaluation of the structures under the higher risk category. Per IBC 2021 §3408.4 (Change of Occupancy), where a change in occupancy results in an increase in risk category, the existing structure must be shown to comply with current code provisions for the new category.

Risk Category

- Current Use (Office): Group B → Risk Category II
- Proposed Use (School): Group E → Risk Category III (per IBC 2021 Table 1604.5)
    - Schools and day-care facilities through grade 12 are designated as Risk Category III.

Seismic Implications

- Importance Factor (Ie):
    - Risk Category II → Ie = 1.0
    - Risk Category III → Ie = 1.25
- This increases seismic design forces by 25% relative to the original office design.
- In Portland's Seismic Design Category D, this would likely necessitate upgrades to lateral systems (plywood shear walls, steel moment frames, diaphragms, and podium shear walls).

Wind & Snow Implications

- Wind Importance Factor:
    - Risk Cat II → 1.0
    - Risk Cat III → 1.15
- Snow Importance Factor:
    - Risk Cat II → 1.0
    - Risk Cat III → 1.1
- These provisions increase roof design loads, requiring verification of mass timber roof framing and diaphragm capacity.

**Summary**

Conversion of the Karuna II buildings from office (Risk Category II) to school (Risk Category III) would increase the risk category under IBC 2021 Table 1604.5. This triggers higher design forces for seismic, wind, and snow, as well as stricter life safety provisions. Structural upgrades to lateral and gravity systems would likely be required for compliance under the current code.

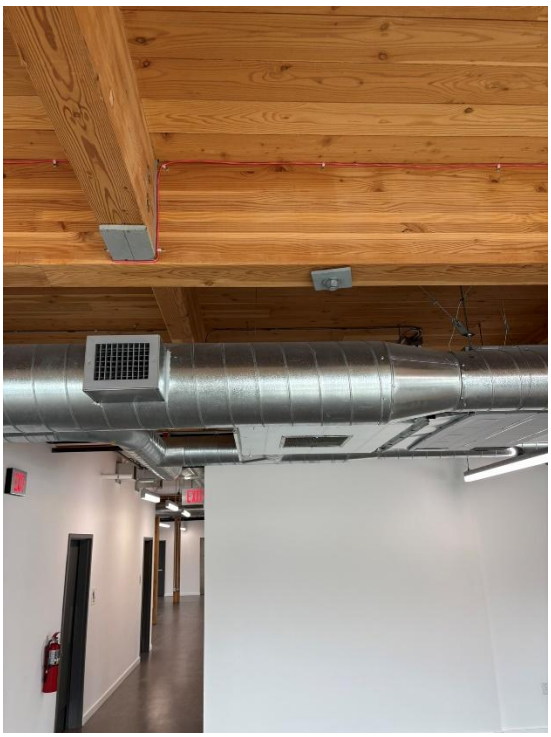
**END OF STRUCTURAL SECTION**

# Structural Observation Photos Karuna II West

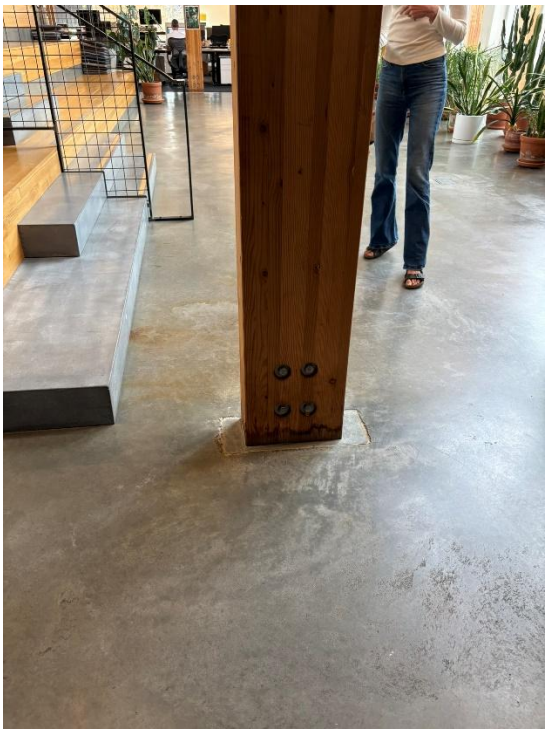


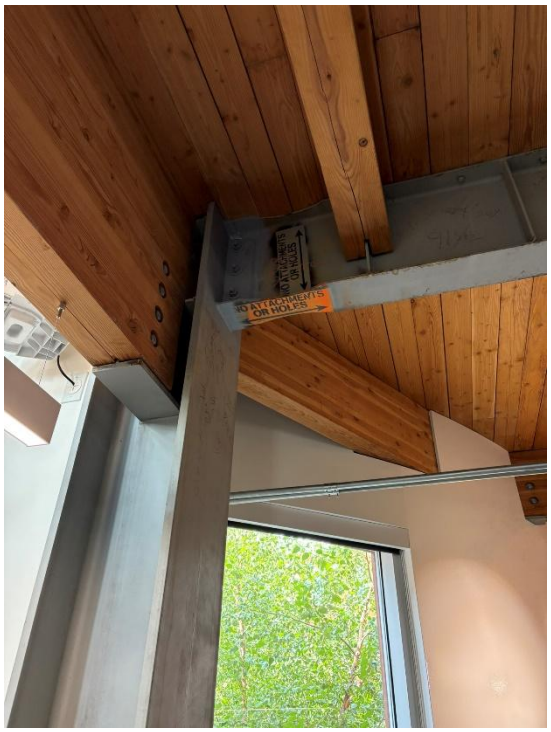


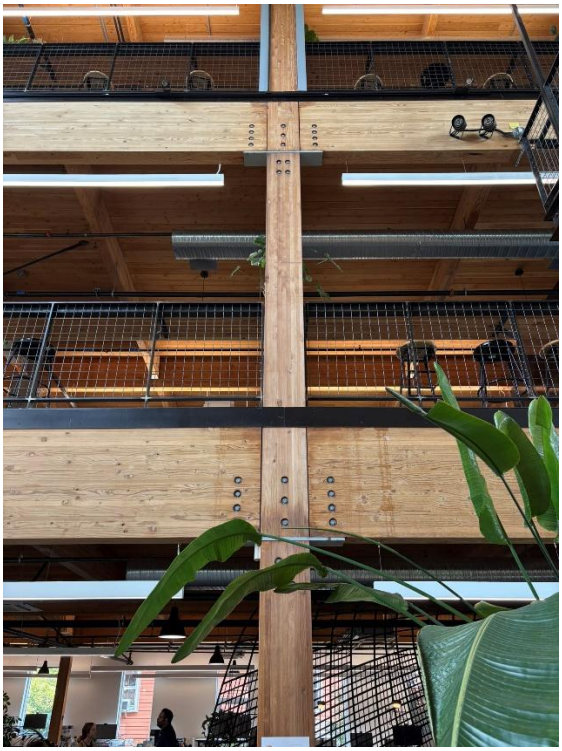




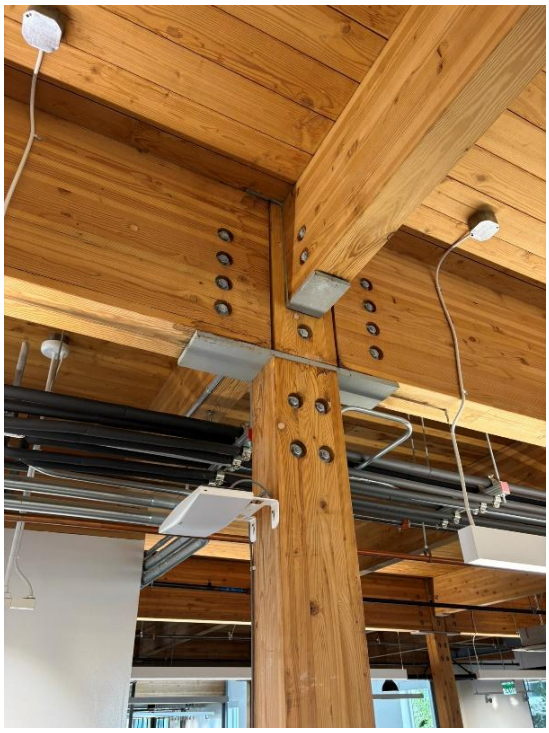
# Structural Observation Photos Karuna II East

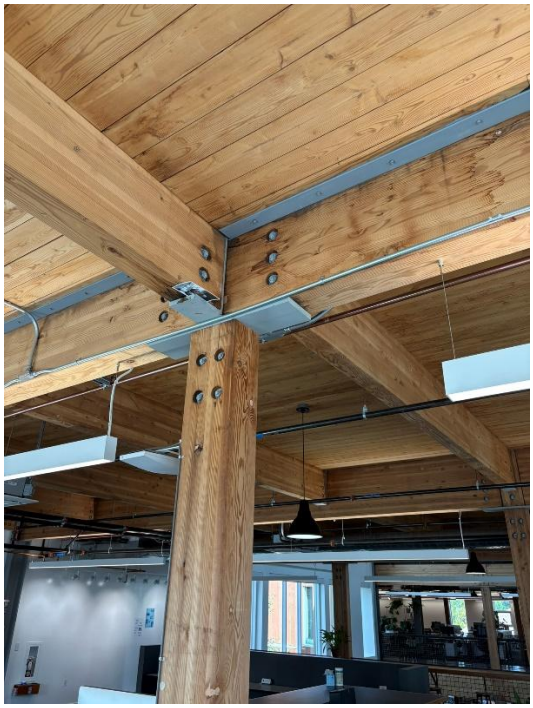


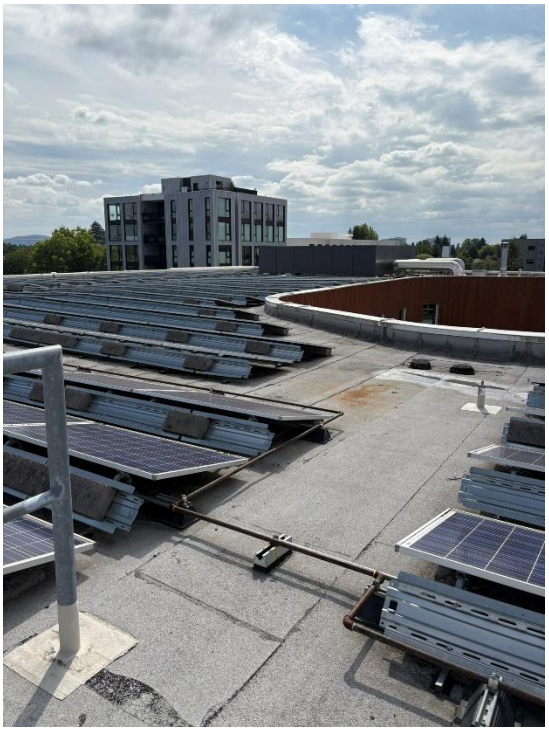












# Mechanical

## Summary of recommended required mechanical cost items

### Heating & Cooling

- VRF system components are in good to excellent condition, but service history is unknown. Interface recommends all coils be cleaned, refrigerant charges checked, and functionality of system components verified.
- Verify ASHRAE 15 requirements are met for existing VRF equipment. ASHRAE 15 requires that refrigerant densities be below a set threshold, to ensure that no space becomes hazardous in the event of a leak in refrigerant piping. Refrigerant charge in existing units is significant and a leak at a branch controller could allow significant refrigerant into enclosed spaces. If refrigerant density proves to be above the Code threshold, this may require the addition of transfer fans.

### Ventilation & Exhaust

- (West Building) Investigate DOAS unit at the roof (HRU-R-01) metallic clicking noise, unit component causing noise should be repaired/replaced.
- (East Building) Level 1 Retail 1 tenant's (restaurant) grease exhaust system access doors for the vertical grease duct risers could not be located during site visit. These must be added if not present to meet NFPA 96 requirements.
- DOAS system's service history is unknown. Interface recommends all coils be cleaned, refrigerant charges checked, fan balancing be checked, and functionality of system components verified.

## Summary of recommended but not required mechanical cost items

### Heating & Cooling

- Copper condensate piping was found to be uninsulated throughout the building. This could lead to condensate forming on piping and dripping in space during cooling mode. No mention was made of this being a past issue during the site visit, however it should be considered.

### Ventilation & Exhaust

- (East Building) If the Level 4 woodburning fireplace is to remain, it poses a potential hazard to occupants who are not familiar with its operation. If the flue is not opened, carbon monoxide could fill the occupied space. Interface does not recommend its use be continued unless training is provided for all future occupants. If the fireplace is removed, its associated flue must be removed as well.

# Mechanical Systems

## Existing System Summary

### Heating & Cooling – Variable Refrigerant Flow (VRF)

Heating and cooling for Karuna II West and Karuna II East is provided by rooftop variable refrigerant flow (VRF) systems with integral heat recovery capability. Systems are isolated to each building and there is no communication of systems between buildings. Systems consist of the following components:

- Outdoor VRF heat pumps: Units are LG branded and are located in screened areas at the roof.
- Refrigerant piping is routed from outdoor heat pumps at the roof to branch controllers within the buildings. Refrigerant piping is routed from each outdoor heat pump to a shaft routed down vertically through the building. Piping penetrates the shaft at each floor and connects to branch controllers.
- Branch controllers: Distributed refrigerant piping routes from each shaft to a branch controller on each level. In most cases branch controllers are located in bathrooms. Refrigerant piping is routed from each branch controller to its associated fan coil.
- Fan coils: In most cases, fan coils are ceiling cassette type, located within the zone they serve. Each contains a fan and heating/cooling coil, which heats/cools the space to maintain the thermostat setpoint.
- Refrigerant piping: The VRF system is based on R-410A refrigerant. must be routed in a rated shaft in accordance with Code requirements for type A2L refrigerants. Where required by Code, shafts are each to be provided with a 4-inch round permanent opening to the exterior, as well as a 4-inch round duct from the bottom of the shaft to the exterior, in accordance with Code requirements for A2L refrigerants.
- Controls: LG branded controls are provided, which are proprietary and not compatible with other VRF manufacturers' equipment.
- Gas usage: Systems are all-electric heat pumps and do not require the use of fossil fuel for their operation.

### Ventilation & Exhaust – Dedicated Outdoor Air System (DOAS) with Heat Recovery

Ventilation air supply for Karuna II West and Karuna II East is provided by rooftop Dedicated Outdoor Air Systems (DOAS) that provide ventilation air supply to each building at a constant airflow. These rooftop units draws fresh ventilation air in and heat/cool it, before supplying to each occupied space via distributed ductwork.

These same DOAS units provide an exhaust airflow that is approximately equal to supply. Exhaust ductwork is routed from the unit at the roof to each space requiring exhaust. Spaces requiring exhaust are primarily bathrooms, however general space exhaust is provided as well to ensure the desired building pressurization is maintained.

DOAS unit cooling is all-electric and does not require the use of fossil fuels when it is in cooling mode. Heating is primarily accomplished using the all-electric heat pump operation of the system, which also does not require the use of fossil fuels when in heating mode. The unit is equipped with an auxiliary gas heating section, which does require the use of fossil fuels. Based on the as-built information provided, this gas heating section is provided for redundancy, to act as a means of backup during periods of extreme cold (when the heat pump heating mode is locked out).

# Discussion West Building "Karuna II West"

## Existing Equipment Information

Existing equipment information is noted below.

### Heating & Cooling

- VRF System Outdoor Heat Pumps
    - HP-R-01, HP-R-02
        - Manufacturer: LG model ARUB288DTE4
        - Location: Roof
        - Year of manufacture: 2015 (approximate)
        - Refrigerant type: R-410A
        - Electrical connection requirement: 460V/3 phase, 60 Hz
        - Total quantity of this unit: 2
- Branch Controllers
    - HRB-01-2, HRB-02-1, HRB-03-1, HRB-04-1, HRB-05-1
        - Manufacturer: LG model PRHR041A
        - Location: Located in the ceiling space at each level, over Toilet Wx02
        - Year of manufacture: 2015 (approximate)
        - Refrigerant type: R-410A
        - Electrical connection requirement: 208V/1 phase, 60 Hz
        - Total quantity of this unit: 4
    - HRB-01-1, HRB-02-2, HRB-03-2, HRB-04-2, HRB-05-2
        - Manufacturer: LG model PRHR041A
        - Location: Located in the ceiling space at each level, over Toilet Wx03
        - Year of manufacture: 2015 (approximate)
        - Refrigerant type: R-410A
        - Electrical connection requirement: 208V/1 phase, 60 Hz
        - Total quantity of this unit: 4
- Fan Coils
    - FC-x
        - Manufacturer: LG. Model number varies based on capacity provided. A majority are ductless ceiling cassette type fan coils, recirculating air in the space and heating/cooling it to meet the thermostat temperature setpoint. Select fan coils (primarily at Level 1) are ducted type, with supply/return ductwork provided for each.
        - Location: In the ceiling space of the zone served
        - Year of manufacture: 2015 (approximate)
        - Refrigerant type: R-410A
        - Electrical connection requirement: 208V/1 phase, 60 Hz
        - Total quantity of this unit: 40. Quantity is estimated, as drawings for some tenant improvements are unavailable.

**Ventilation & Exhaust**

- Dedicated Outside Air System (DOAS)
    - HRU-R-01
        - Manufacturer: Aaon model RN-016
        - Area served: Whole building
        - Location: Roof
        - Year of manufacture: 2015 (approximate)
        - Refrigerant type: R-410A
        - Electrical connection requirement: 460V/3 phase, 60 Hz
        - Total quantity of this unit: 1
- Exhaust Only Systems
    - EF-01-1 (Electrical Room)
        - Manufacturer: Greenheck model 250
        - Year of manufacture: 2015 (approximate)
        - Electrical connection requirement: 120V/1 phase, 60 Hz
        - Total quantity of this unit: 1
    - EF-01-2 (Trash Room)
        - Manufacturer: Greenheck model SP-A250
        - Year of manufacture: 2015 (approximate)
        - Electrical connection requirement: 120V/1 phase, 60 Hz
        - Total quantity of this unit: 1
- Distribution ductwork
    - Distribution ductwork for ventilation air supply and exhaust is uninsulated spiral ductwork. This is in compliance with Code for the installation provided, with the ductwork exposed in the occupied areas.

### Controls

The building is provided with LG branded VRF system controls. These are proprietary and any modified systems will require the continued usage of LG branded controls.

# Discussion East Building "Karuna II East"

## Existing Equipment Information

Existing equipment information is noted below.

### Heating & Cooling

- VRF System Outdoor Heat Pumps
    - HP-R-01, HP-R-02
        - Manufacturer: LG model ARUB288DTE4
        - Location: Roof
        - Year of manufacture: 2014 (approximate)
        - Refrigerant type: R-410A
        - Electrical connection requirement: 460V/3 phase, 60 Hz
        - Total quantity of this unit: 2
- Branch Controllers
    - HRB-01-1, HRB-02-1, HRB-03-1, HRB-04-1
        - Manufacturer: LG model PRHR041A
        - Location: Located in the ceiling space at each level, directly to the east of Stair S2
        - Year of manufacture: 2014 (approximate)
        - Refrigerant type: R-410A
        - Electrical connection requirement: 208V/1 phase, 60 Hz
        - Total quantity of this unit: 4
    - HRB-01-2, HRB-02-2, HRB-03-2, HRB-04-2
        - Manufacturer: LG model PRHR041A
        - Location: Located in the ceiling space at each level, over the north restrooms
        - Year of manufacture: 2014 (approximate)
        - Refrigerant type: R-410A
        - Electrical connection requirement: 208V/1 phase, 60 Hz
        - Total quantity of this unit: 4
- Fan Coils
    - FC-x
        - Manufacturer: LG. Model number varies based on capacity provided. All are ductless ceiling cassette type fan coils, recirculating air in the space and heating/cooling it to meet the thermostat temperature setpoint.
        - Location: In the ceiling space of the zone served
        - Year of manufacture: 2015 (approximate)
        - Refrigerant type: R-410A
        - Electrical connection requirement: 208V/1 phase, 60 Hz
        - Total quantity of this unit: 26

### Ventilation & Exhaust

- Dedicated Outside Air System (DOAS)
    - HRU-R-01
        - Manufacturer: Aaon model RN-016
        - Area served: Whole building
        - Location: Roof
        - Year of manufacture: 2014 (approximate)
        - Refrigerant type: R-410A
        - Electrical connection requirement: 460V/3 phase, 60 Hz
        - Total quantity of this unit: 1
- Supply/Transfer Air Systems
    - SF-1 through SF-10
        - Manufacturer: Greenheck model SQ
        - Area served: Conference rooms
        - Location: Ceiling adjacent to conference rooms
        - Year of manufacture: 2015 (approximate)
        - Electrical connection requirement: 115V/1 phase, 60 Hz
        - Total quantity of this unit: 10
    - Level 1 Retail 1 restaurant tenant makeup air unit
        - Manufacturer: Greenheck model DGK-109-H15-01-HZ
        - Area served: Level 1 Retail 1 restaurant tenant
        - Year of manufacturer: Unknown
        - Location: Roof
        - Electrical Connection requirement: 208V/3 phase, 60 Hz
        - Total quantity of this unit: 1
        - Ductwork was not visible at time of site visit aside from that exposed at the roof. Based on the unit location and ductwork at the roof, it appears to route vertically on the north side of Stair 2.
- Exhaust Only Systems
    - EF-01-1 (Recycle Room)
        - Manufacturer: Cook
        - Year of manufacture: 2014 (approximate)
        - Electrical connection requirement: 120V/1 phase, 60 Hz
        - Total quantity of this unit: 1
    - EF-01-2 (Electrical Room)
        - Manufacturer: Cook
        - Year of manufacture: 2014 (approximate)
        - Electrical connection requirement: 120V/1 phase, 60 Hz
        - Total quantity of this unit: 1
    - Fireplace Flue
        - Level 4 contains a woodburning fireplace, with a manually operated flue that discharges vertically to the roof.

  - HD-1 range exhaust hood
    - Manufacturer: Greenheck model GO-48.00-S
    - Year of manufacturer: 2015 (approximate)
    - Location: Break area kitchen, directly over gas range
    - Electrical connection requirement: Unknown. Hoods of this type are most commonly 120V/1 phase.
    - Total quantity of this unit: 1
  - EF-1 (Serves HD-1 range exhaust hood)
    - Manufacturer: Greenheck model CUE-099-A-G
    - Year of manufacturer: 2015 (approximate)
    - Location: Roof, directly over break area kitchen
    - Electrical connection requirement: Unknown
    - Total quantity of this unit: 1
  - Level 1 Retail 1 restaurant tenant exhaust
    - Manufacturer: Greenheck model USF-315-10-BI-7-G
    - Area served: Level 1 Retail 1 restaurant tenant grease exhaust hood
    - Year of manufacturer: Unknown
    - Location: Roof
    - Electrical Connection requirement: 208V/3 phase, 60 Hz
    - Total quantity of this unit: 1
    - Ductwork was not visible at time of site visit aside from that exposed at the roof. Based on the unit location and ductwork at the roof, it appears to route vertically on the north side of Stair 2.

- Distribution ductwork
  - Distribution ductwork for ventilation air supply and exhaust is uninsulated spiral ductwork. This is in compliance with Code for the installation provided, with the ductwork exposed in the occupied areas.

#### Controls

The building is provided with LG branded VRF system controls. These are proprietary and any modified systems will require the continued usage of LG branded controls.

# Recommendations

## Karuna II West

#### Heating & Cooling

- VRF system components are in good to excellent condition. The existing VRF system capacity is likely sufficient to meet the future heating and cooling needs, provided the new space usage is similar to that of the existing tenants. At an age of approximately 11 years, their anticipated remaining life is 9-14 years. At the end of this equipment life, all components associated with the VRF system will likely require replacement. This includes the refrigerant piping routed within rated shafts, as the existing system's refrigerant is currently being phased out. While refrigerant

will be available for unit servicing, a full system replacement will use an alternate refrigerant that has a lower Global Warming Potential.
- A revised space program will require the removal/relocation/addition of select VRF fan coils. When fan coils are relocated/added, their associated refrigerant piping, power, and condensate connections will require modification as well. This will require evacuating the refrigerant in the existing system, installing new refrigerant piping, and replacing the refrigerant in the system.
- Caution should be taken by the Engineer of Record for the revised space program not to locate VRF branch controllers or fan coils in excessively small spaces. ASHRAE 15 requires that refrigerant densities be below a set threshold, to ensure that no spaces become hazardous in the event of a leak in refrigerant piping.
- Ventilation air supply ductwork is currently uninsulated in the occupied space. This is Energy Code compliant in its current installation. If ceilings are added to the space in the future, insulation will need to be added to the existing ductwork.

#### Ventilation & Exhaust

- The DOAS unit at the roof (HRU-R-01) is in fair to good condition. An abnormal noise was observed at the unit during the site visit and should be investigated to determine its source. The noise was a metallic clicking noise within the unit, which was audible at 5-10 second intervals. At an age of approximately 10 years, its anticipated remaining life is 10 years. At the end of this equipment life, the unit will likely require replacement.

- EF-01-1 and EF-01-2 are in good condition. At an age of approximately 10 years, their anticipated remaining life is 10-15 years. At the end of this equipment life, the unit will likely require replacement.

## Karuna II East

#### Heating & Cooling

- VRF system components are in good to excellent condition. The existing VRF system capacity is likely sufficient to meet the future heating and cooling needs, provided the new space usage is similar to that of the existing tenants. At an age of approximately 11 years, their anticipated remaining life is 9-14 years. At the end of this equipment life, all components associated with the VRF system will likely require replacement. This includes the refrigerant piping routed within rated shafts, as the existing system's refrigerant is currently being phased out. While refrigerant will be available for unit servicing, a full system replacement will use an alternate refrigerant that has a lower Global Warming Potential.
- A revised space program will require the removal/relocation/addition of select VRF fan coils. When fan coils are relocated/added, their associated refrigerant piping, power, and condensate connections will require modification as well. This will require evacuating the refrigerant in the existing system, installing new refrigerant piping, and replacing the refrigerant in the system.
- Caution should be taken by the Engineer of Record for the revised space program not to locate VRF branch controllers or fan coils in excessively small spaces. ASHRAE 15 requires that refrigerant densities be below a set threshold, to ensure that no space becomes hazardous in the event of a leak in refrigerant piping.

- Ventilation air supply ductwork is currently uninsulated in the occupied space. This is Energy Code compliant in its current installation. If ceilings are added to the space in the future, insulation will need to be added to the existing ductwork.

### Ventilation & Exhaust

- The DOAS unit at the roof (HRU-R-01) is in good condition. At an age of approximately 10 years, its anticipated remaining life is 10 years. At the end of this equipment life, the unit will likely require replacement.
- SF-1 through SF-10 provide ventilation air to conference rooms by circulating air from the adjacent open office into each conference room. This current installation is not Code compliant, and Interface recommends DOAS distribution ductwork be modified to supply ventilation air directly into any enclosed offices.
- EF-01-1 and EF-01-2 are in good condition. At an age of approximately 10 years, their anticipated remaining life is 10-15 years. At the end of this equipment life, the unit will likely require replacement.
- The Level 4 woodburning fireplace poses a potential hazard to occupants who are not familiar with its operation. If the flue is not opened, carbon monoxide could fill the occupied space. Interface does not recommend its use be continued unless training is provided for all future occupants.
- The existing Level 1 Retail 1 tenant's (restaurant) grease exhaust system was not visible at the time of the site visit, aside from the fan and ductwork at the roof. Access doors for the vertical grease duct riser could not be located and must be added if not present, in accordance with NFPA 96 requirements.
- HD-1 range exhaust hood serving Level 1 is in excellent condition. No modifications are required to the hood if the range below is left in place.
- EF-1 is in good condition. At an age of approximately 10 years, its anticipated remaining life is 10-15 years. At the end of this equipment life, the unit will likely require replacement.
- The Level 1 Retail 1 tenant makeup air unit is in good condition. Age is unknown, but based on appearance, appears to be approximately 10 years. Based on this its anticipated remaining life is 10 years. At the end of this equipment life, the unit will likely require replacement.
- The Level 1 Retail 1 tenant exhaust fan is in good condition. Age is unknown, but based on appearance, appears to be approximately 10 years. Based on this its anticipated remaining life is 10-15 years. At the end of this equipment life, the unit will likely require replacement.

**END OF MECHANICAL SECTION**

# Plumbing

## Summary of recommended required Plumbing cost items

- Scope the existing sanitary system to verify condition, location (as built), invert elevation.
- Provide and install missing roof/overflow drain grates that are missing.

## Summary of recommended but not required Plumbing cost items

- Cleaning primary roof drains.
- Consider revising hot water system in building
- Consider replacing corded roof gas piping.

## Plumbing Systems

### Drainage Waste and Vent, Sanitary Sewer System

- The sanitary system is original to the building. We expect the sanitary system to be cast-iron below grade.

### Domestic hot, cold-water systems

- Domestic water piping system is copper and is original to the building, appears to be in good working condition.
- The 3-inch domestic water backflow is located in each building's mechanical room.
- Existing gas fired hot water heater is a 100-gallon located in each building's mechanical room with hot water recirculation pump. The gas load is 199 CFH and serves the entire building. On the third-floor water was ran for over a minute and no hot water was produced. We recommend replacing this gas fired water heater with electric tank type system or a heat pump version with a storage tank and demo gas piping back to the meter.
- (East Building) Breakroom sink is supplied with under counter instantaneous Eemax water heater. Recommended to remove and route HW from main building system to sink.

### Natural Gas System

- Meter bank on the exterior of the buildings with three gas meters at East Building and one gas meter at West Building installed and room for additional meters as needed.
- Gas piping on roof is not painted and rusted. We recommend replacing gas piping and painting piping to keep from erosion.

## Plumbing fixtures

- The plumbing fixtures appear to be in good condition.
- The water closets are ADA complaint, elongated bowl, open seat and are flush valve type.
- Urinal is wall hung ADA complaint with hard wired sensor flush valve type.
- Lavatory are wall mounted is ADA complaint with some having sensor hard wired faucets and others manual operated single handle faucets. Hot water to lavatories is provided by electrical tankless water heaters.
- (East Building) Breakroom sinks appear to be in fair condition, with manual faucet and appear to be ADA compliant. Seems to be supplied with cold water only.
- (West Building) Two shower stalls one ADA stall with pull down seat and one standard shower. Both in good condition.

**END OF PLUMBING SECTION**

# Electrical

## Summary of recommended required Electrical cost items

- In public spaces, replace existing receptacles in place with tamper-proof receptacles.
- Relocate interior panels as needed to match new building layout.
- Investigate condition of luminaires, power circuit, or control circuit in East Building Stair 1 and repair/replace if needed.

## Summary of recommended But Not required Electrical cost items

- Consider relocating existing trapeze-mounted transformer in West Building Men's restroom on the third floor.
- Consider a maintenance and cleaning plan for PV arrays on both buildings.
- Consider reusing existing lighting throughout both buildings.
- Consider replacing existing cracked receptacle faceplates.
- Consider replacing the main multi-meter switchboard or a portion of the switchboard for a single metered service serving the PPS space.

## Discussion - Site

Power to both buildings is provided by a 500 kVA, 12470Y/7200-480Y/277V pad-mounted transformer served by Pacific Power. The transformer is in a gated enclosure and appears to be in good condition. The transformer also serves the building to the northwest.




*Figure E-70 - Pacific Power Transformer*

In addition to building mounted lights, there are LED bollards, both tall and short, placed in the landscaping throughout the interior courtyard to provide general lighting.



*Figure E-71 - Area Bollards*

GFCI receptacles with waterproof covers are also scattered throughout the landscaping for convenience.

# Discussion West Building "Karuna II West"

## Main Service

The site Pacific Power transformer serves an 800A, 480/277V, 3-phase, 4-wire Siemens commercial meter center switchboard 'MDP-4A1-1', located in the building main electrical room on the main floor. There is an 800A main disconnect. 800A is more than adequate for building needs and should also be capable of electrification transition off of use of natural gas. The switchboard has spaces for (12) 200A disconnects with utility meters or space for utility meters. The following utility meters and disconnects are currently in use. All disconnects are 200A unless otherwise noted.

1. Spare
2. Spare
3. Spare
4. Spare
5. Panel 4A3-1 – [redacted]
6. Disconnect T2A1-2 – [redacted]
7. Panel 4A1-2 – [redacted]
8. Spare
9. Spare
10. Panel 4A2-1 – [redacted]
11. 4A5-1 – [redacted]
12. Panel 4A1-1 – Karuna II West House Panel – 1st Floor

The power company typically requires separate addresses assigned to building remain to keep a multi-meter center arrangement. PPS could keep this arrangement with separate meters; however, it could consider replacing the switchboard, or a portion of the switchboard, with a standard main service switchboard or larger service with distribution section or panel sized for the PPS space.

## General Conditions

480Y/277V and 208Y/120V distribution panelboards are located on each floor, installed by the tenants. Level 3 panels were installed at the time of original construction and serve all tenant spaces on the floor. Existing panelboards are in good condition. Reuse and relocate as needed. Extend existing feeders and circuits to be saved as needed.

- 1st Floor Main Electrical Room (House Panels):
    - 480Y/277V, 3-phase, 4-wire, 200A panelboard '4A1-1'
    - 75 kVA transformer 'XFRM-T2A1-1'
    - 2-section 208Y/120V, 3-phase, 4-wire, 200A panelboard '2A1-1' and '2A1-2' fed from 'T2A1-1'
    - 480Y/277V, 3-phase, 4-wire, 60A panelboard 'S4A1-1', fed from 70A standby automatic transfer switch 'ATS-S1'.

- Level 2:
    - 480Y/277V, 3-phase, 4-wire, 200A panelboard '4A2-1' fed from main switchboard
    - 75 kVA transformer 'T2A2-1'
    - 2-section 208Y/120V, 3-phase, 4-wire, 200A panelboard '2A2-1' and '2A2-2' fed from 'T2A2-1'
- Level 3 (House Panels):
    - 480Y/277 V, 3-phase, 4-wire, 200A panelboard '4A3-1' fed from main switchboard
    - 75 kVA transformer 'T2A3-1' (trapeze mounted above the door in the men's restroom)
    - 2-section, 208Y/120V, 3-phase, 4-wire, 200A panelboard '2A3-1' and '2A3-2' fed from 'XFRM-2A3-1'.
- Level 5:
    - 480Y/277 V, 3-phase, 4-wire, 200A panelboard '4A5-1' fed from main switchboard
    - 75 kVA transformer 'T2A5-1'
    - 2-section 208Y/120V, 3-phase, 4-wire, 200A panelboard '2A5-1' and '2B5-2' fed from 'T2B5-1'

Consider relocating the third-floor transformer, currently located above the door in the men's restroom, to a more secure area to reduce risk of damage.



*Figure E-72- 3rd Floor Transformer*

Panels on Level 5 currently feed tenant spaces on Level 4 and Level 5. There is infrastructure in place (spare conduit) from the main electrical room to Level 4 to feed new panels if needed.

A low-voltage energy meter is located in the main electrical room to measure usage.

Recommend providing new receptacle layout per floor plans, reuse existing receptacles and receptacle circuits when available. Per NEC 406.12(4), tamper-resistant receptacles are required in education facilities. While there are some existing receptacles that are tamper-resistant, most should be replaced with new tamper-resistant receptacles.

There are floor receptacles in tenant areas throughout the building. If these are not in a location that can be reused, they will need to be demolished and the floor patched.

There is a baseboard raceway and above-ground raceway to provide power to desks in Suite 310 (Figure E-73). Consider removing this if space is to be reused.

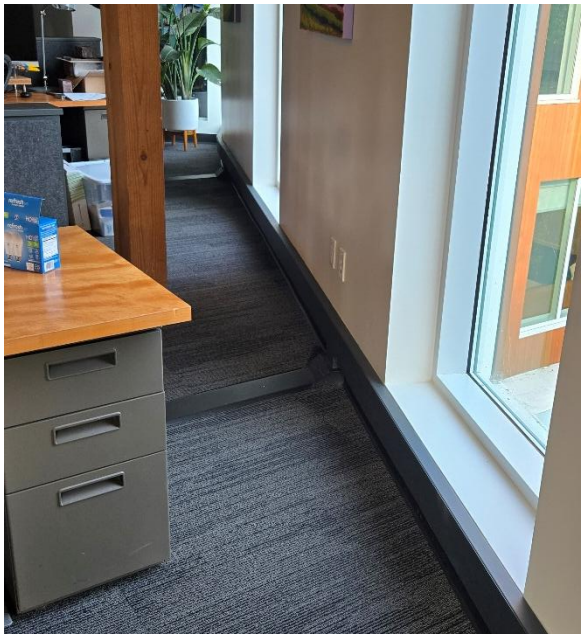


*Figure E-73 - Suite 310 Floor Raceway*

## PV Array

There is a 30 kW PV array on the roof of the building. A solar DC disconnect is located on the roof, and a solar AC disconnect is located in the main electrical room. The PV array appears to be in good condition, however debris is building up at the bottom portion of the panels and should be cleaned. Consider a maintenance and cleaning plan for the PV arrays.

The PV array serves panel '4A1', a house panel located in the main electrical room.

## Emergency Power

There is a 35kW/45kVA, 480Y/277V, 3-phase, 3-wire natural gas generator located on the roof. This feeds panel 'S4A1-1' via standby automatic transfer switch 'ATS-S1', located in the first-floor main electrical room. This serves the elevator. Note that this elevator is required to be on standby power because the

west building is 5 stories. At 4 stories, the elevator in the east building is not required to be on standby power.

Emergency lights and exit signs are served via a series of lighting inverters located in the main electrical room and throughout the building. The City of Portland does not allow emergency power to egress lights to be served by a natural gas generator due to seismic risk, thus the existing generator cannot serve emergency lights.

## Fire Alarm

Building is served by a Simplex 4010ES addressable fire detection and control system manufacturer by SimplexGrinnell, now part of Johnson Controls. The communication protocols are exclusive to Simplex systems and are not compatible with devices from other manufacturers without specific integration modules or gateways. The 4010ES supports 1,000 addressable points.

## Lighting

Existing lights throughout the building are LED and in good condition. Existing lighting appears to be in good condition, recommend to reuse as much as possible. Different lighting strategies are used in different tenant spaces, if consistent lighting is desired throughout the building, consider new lights or reusing and relocating light fixtures.

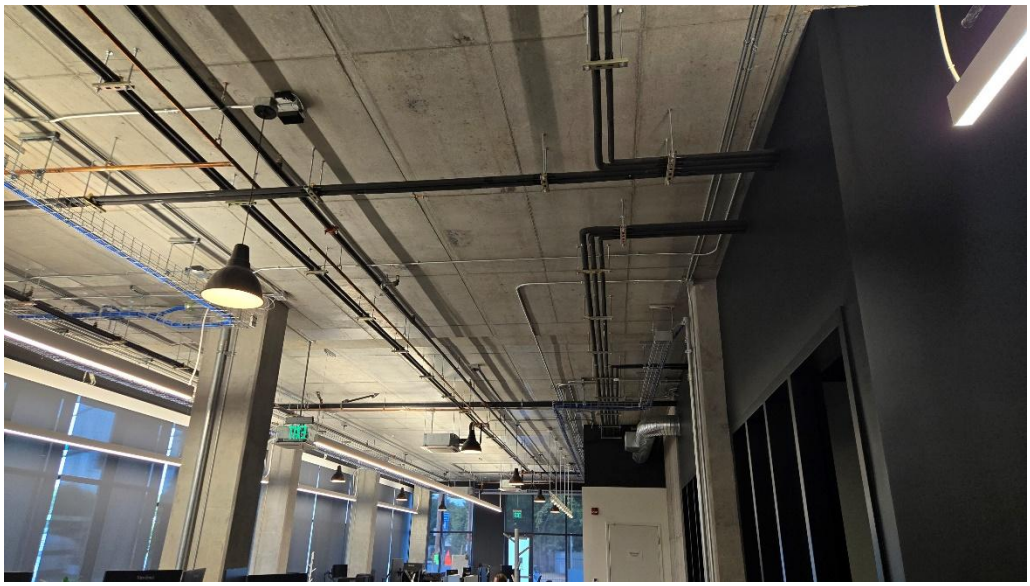


*Figure E-74- 1st Floor Tenant Lighting*

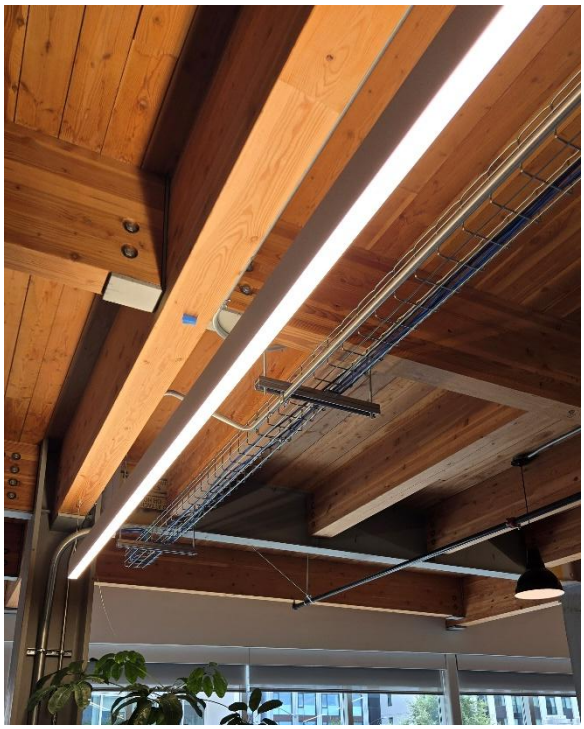


*Figure E-75- 2nd Floor Tenant Lighting*

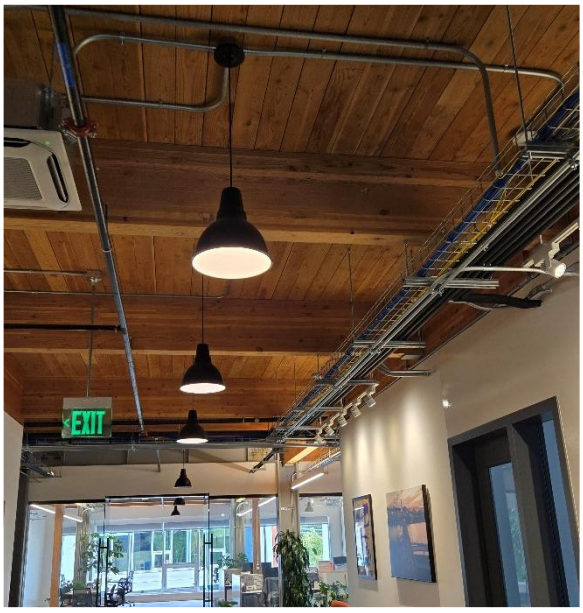



*Figure E-7 - 2nd Floor Tenant Lighting*



*Figure E-76- 3rd Floor Hallway Lighting*

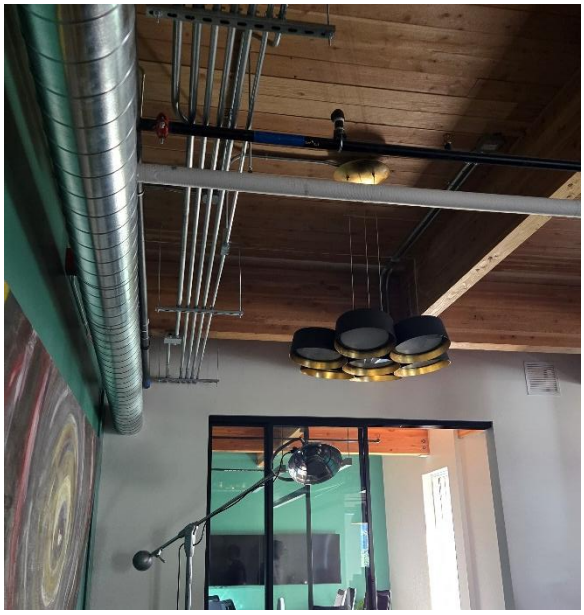


*Figure E-77- Suite 310 Decorative Pendant*

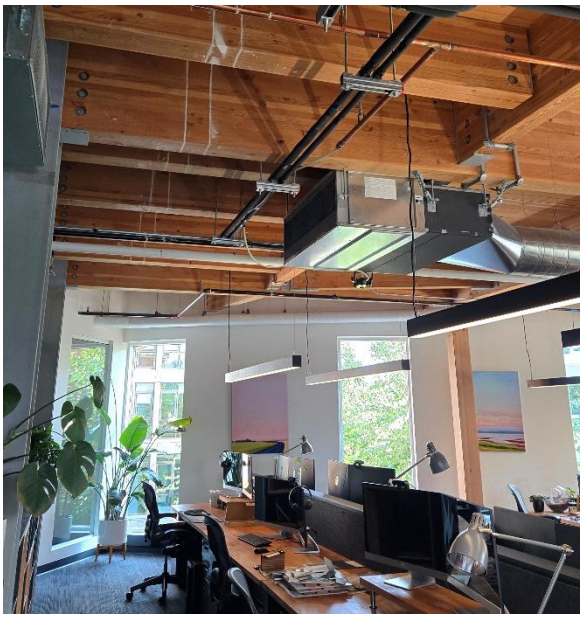


*Figure E-78- Suite 310 Office Lighting*



*Figure E-79 - 5th Floor Tenant Lighting*

# Discussion East Building "Karuna II East"

## Main Service

The site Pacific Power transformer serves an 800A, 480/277V, 3-phase, 4-wire Siemens commercial meter center switchboard 'MDP-4A1-1', located in the building main electrical room on the main floor. There is an 800A main disconnect. 800A is more than adequate for building needs and should also be capable of electrification transition off of use of natural gas. The switchboard has spaces for (12) 200A disconnects with utility meters or space for utility meters. The following utility meters and disconnects are currently in use. All disconnects are 200A unless otherwise noted.

1. Spare
2. Spare
3. Spare
4. XFMR T2B1-2 – [redacted]A)
5. 4B2-1 – [redacted]
6. 4B3-1 – [redacted]
7. 4B1 – Karuna II East
8. [redacted]
9. Spare
10. Spare
11. Spare
12. [redacted]

The power company typically requires separate addresses assigned to building remain to keep a multi-meter center arrangement. PPS could keep this arrangement with separate meters; however, it could consider replacing the switchboard, or a portion of the switchboard, with a standard main service switchboard or larger service with distribution section or panel sized for the PPS space.

## General Conditions

480Y/277V and 208Y/120V distribution panelboards are located on each floor, installed by the tenants. Existing panelboards are in good condition. Reuse and relocate as needed. Extend existing feeders and circuits to be saved as needed.

- 1st Floor Main Electrical Room (House Panels):
    - 480Y/277V, 3-phase, 4-wire, 200A panelboard '4B1-1'
    - 75 kVA transformer 'XFRM-T2B1-1'
    - 2-section 208Y/120V, 3-phase, 4-wire, 200A panelboard '2B1-1' and '2B1-2' fed from 'T2B1-1'
- Level 2:
    - 480Y/277V, 3-phase, 4-wire, 200A panelboard '4B2-1' fed from main switchboard
    - 75 kVA transformer 'T2B2-1'
    - 2-section 208Y/120V, 3-phase, 4-wire, 200A panelboard '2B2-1' and '2B2-2' fed from 'T2B2-1'

- Level 3:
    - 480Y/277 V, 3-phase, 4-wire, 200A panelboard '4B3-1' fed from main switchboard
    - 2-section, 208Y/120V, 3-phase, 4-wire, 200A panelboard '2B3-1' and '2B3-2' fed from panel '2B1-3' in main electrical room
- Level 4:
    - 480Y/277 V, 3-phase, 4-wire, 200A panelboard '4B4-1' fed from '4B3-1'
    - 75 kVA transformer 'T2B4-1'
    - 2-section 208Y/120V, 3-phase, 4-wire, 200A panelboard '2B4-1' and '2B2-4' fed from 'T2B4-1'

A low-voltage energy meter is located in the main electrical room to measure usage.

Recommend providing new receptacle layout per floor plans, reuse existing receptacles and receptacle circuits when available. Per NEC 406.12(4), tamper-resistant receptacles are required in education facilities. While there are some existing receptacles that are tamper-resistant, most should be replaced with new tamper-resistant receptacles.

There are floor receptacles in tenant areas throughout the building. If these are not in a location that can be reused, they will need to be demolished and the floor patched.

Faceplates on some receptacles are cracked. Recommend replacing faceplates as needed.

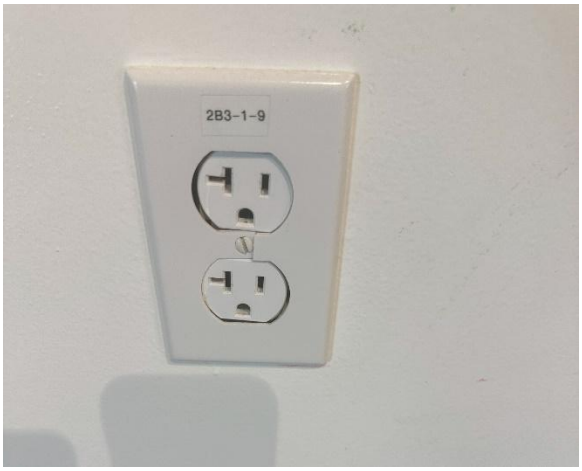



*Figure E-80 - Cracked Receptacle Faceplate*

## PV Array

There are (2) 23 kW PV arrays located on the roof. An inverter and PV disconnect for each array are located in a recessed enclosure in the exterior parking area. DC disconnects are located on the roof. The PV arrays appears to be in good condition, however debris is building up at the bottom portion of the panels and should be cleaned.  Consider a maintenance and cleaning plan for the PV arrays.



*Figure E-81 - PV Array on Roof*




*Figure E-82 - PV Inverter and Disconnect*

The PV array serves panel '4B1-1', a house panel located in the main electrical room.

## Emergency Power

Emergency lights and exit signs are served via a series of lighting inverters located in the main electrical room and throughout the building. There are (3) IOTA inverters located in the main electrical room.

## Fire Alarm

Building is served by a Simplex 4100ES addressable fire detection and control system manufacturer by SimplexGrinnell, now part of Johnson Controls. The communication protocols are exclusive to Simplex systems and are not compatible with devices from other manufacturers without specific integration modules or gateways. The 4100ES supports 3,000 addressable points and includes a system-wide voice notification with UL 2572 listing for mass notification / emergency communications including audio amplification and firefighter telephone integration.

The Level 4 Atrium drop down smoke curtains had some damage. During the site walk, the Owner's Representative mentioned a work order was in progress to get these repaired. The smoke curtains only serve Level 4, separating the office space from the Atrium in the event of a fire.

## Lighting

Existing lights throughout the building are LED and in good condition. Existing lighting appears to be in good condition, recommend to reuse as much as possible.

Lights in Stair 1 were off at time of site visit. Investigate condition of luminaires, power circuit, or control circuit and repair/replace if needed.

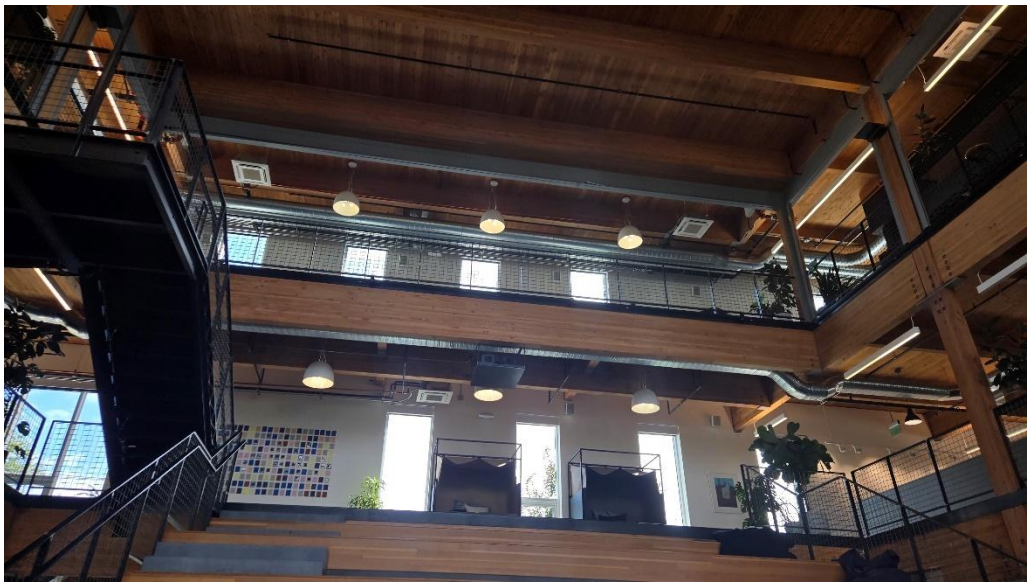


*Figure E-83 - Pendants and Linear pendants throughout tenant space*

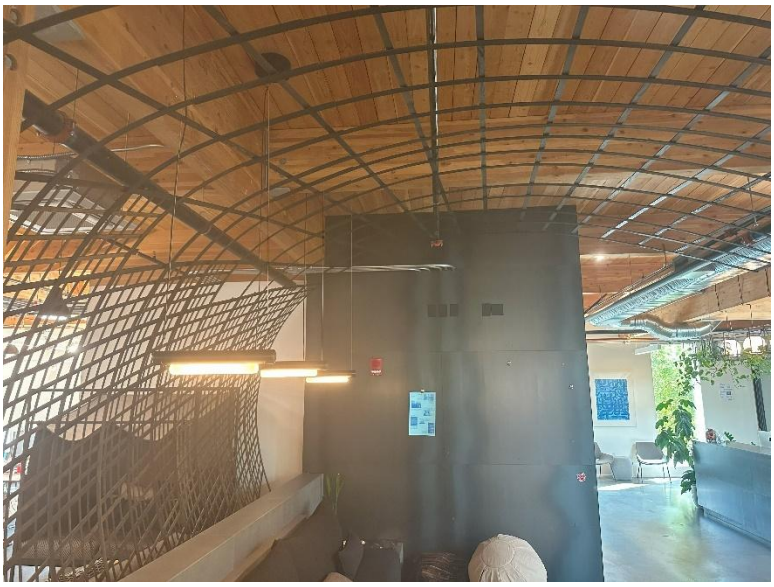


*Figure E-84 - Pendants in tenant lobby*

**END OF ELECTRICAL SECTION**